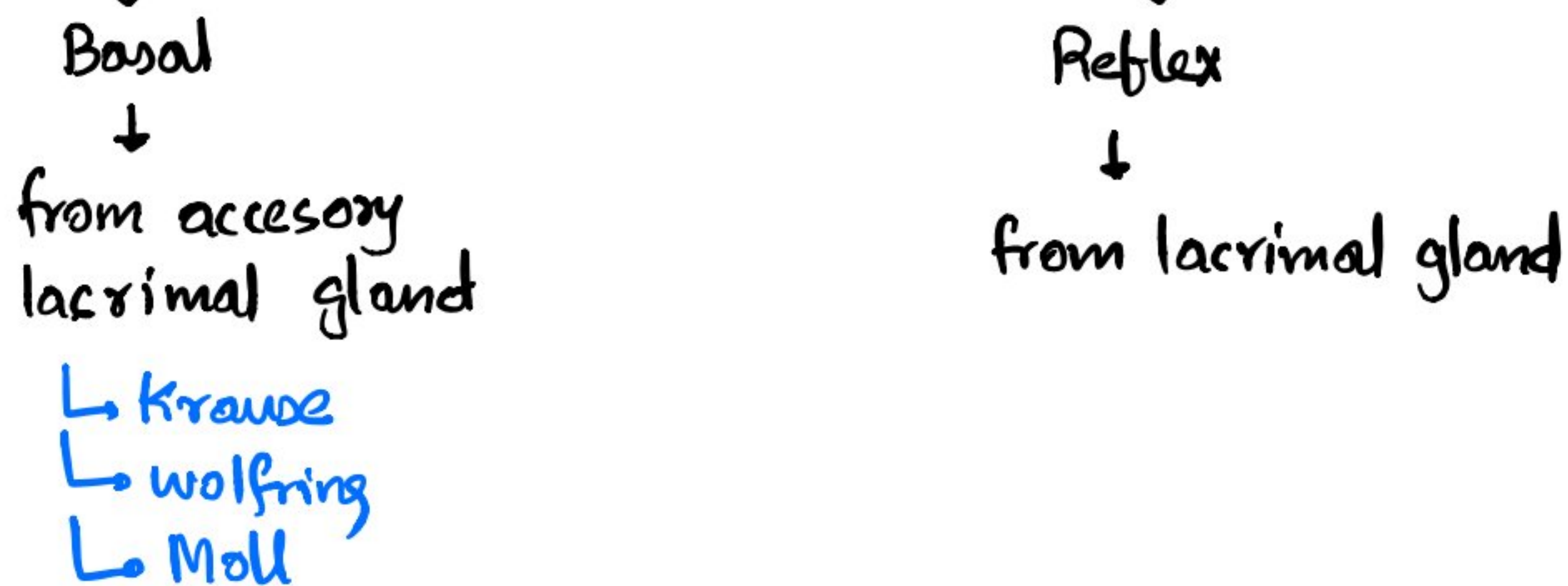


Aqueous layer : Thickest layer

(MIDDLE)



- Lipid layer ∴ Prevent evaporation
(Outer) ∴ secreted by meibomian glands.

Tear film:

pH = 7.4 (Acidic)

R.I = 1.357

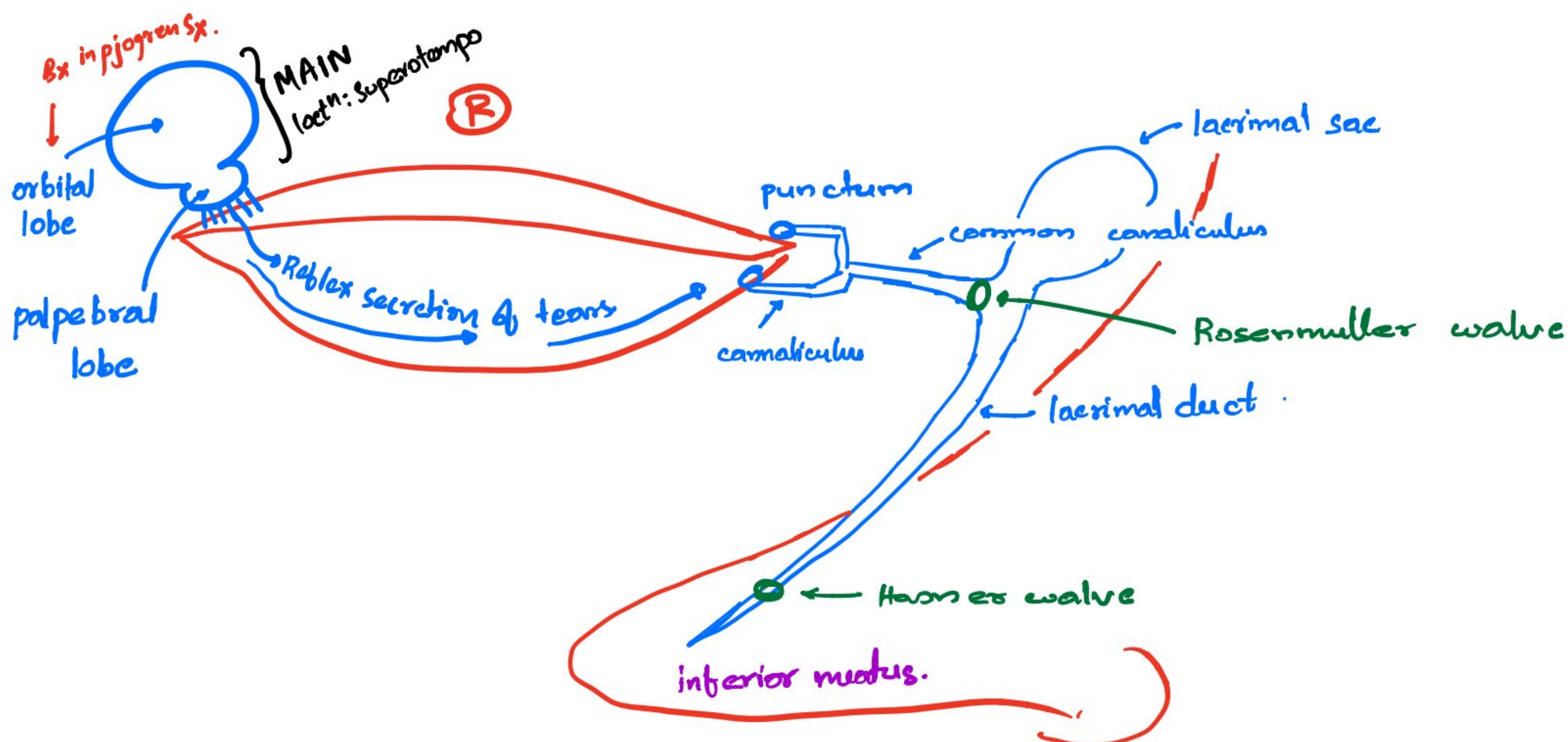
Basal rate of production = 1.2 µl/min

Mucous layer

(INNER)

→ secreted by $\begin{cases} \text{Mamz} \\ \text{goblet cells} \end{cases}$ → $\begin{cases} \text{Max: nasal conjunctiva} \\ \text{Min: Superior conjunctiva} \end{cases}$

Lacrimal Apparatus of Eye:



- Lacrimal sac = 8-12mm
- Length of NLD = 12-18mm

Direction of NLD

• Backward → Lateral → Down

• To open anteriorly to inferior meatus

- For unidirection → Hasner valve
- Rosenmuller valve

• LPS aponeurosis divides main L.G. in:
• Superficial palpebral lobe
• Deep orbital lobe

Nasolacrimal duct obstruction

1) Epiphora: Increased watering from eye

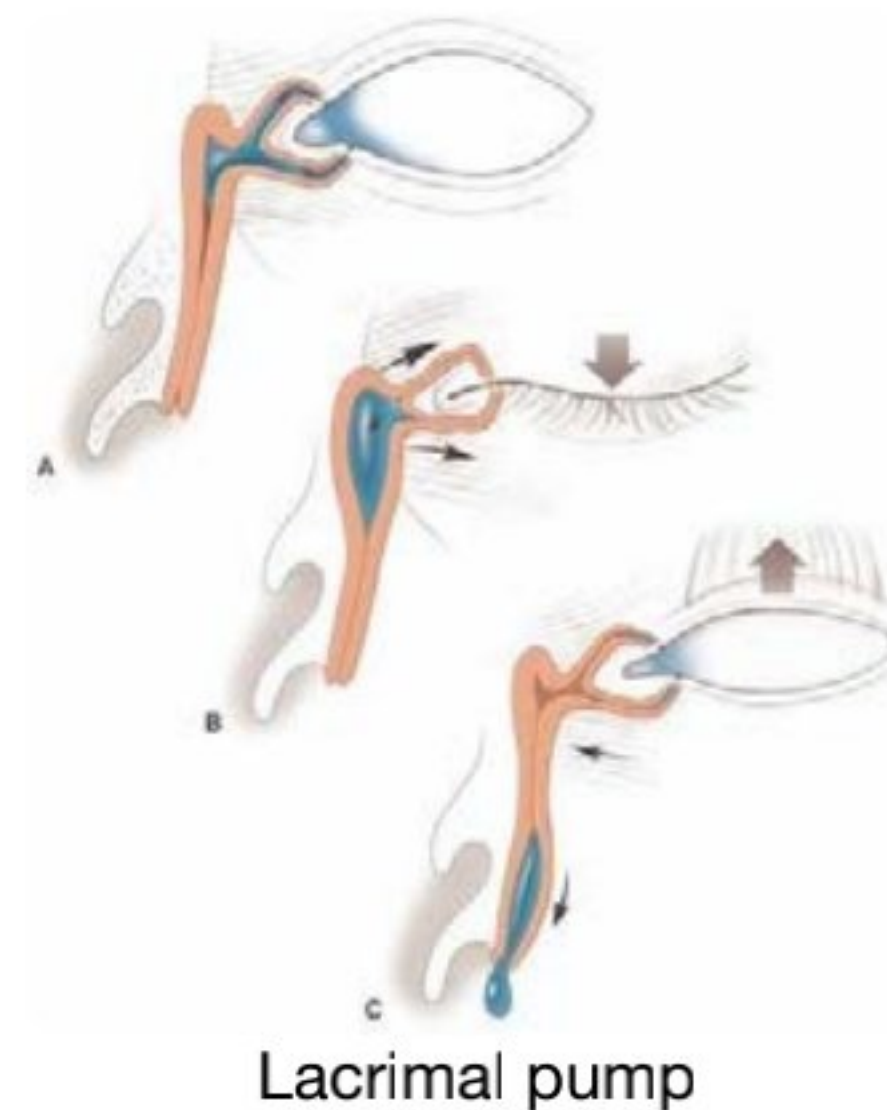
↓
water collects in lacrimal sac

↓
mucus

↓
infection

2) Dacrocystitis → Mucocoele

Regurgitation test +ve & mucous comes out when probed



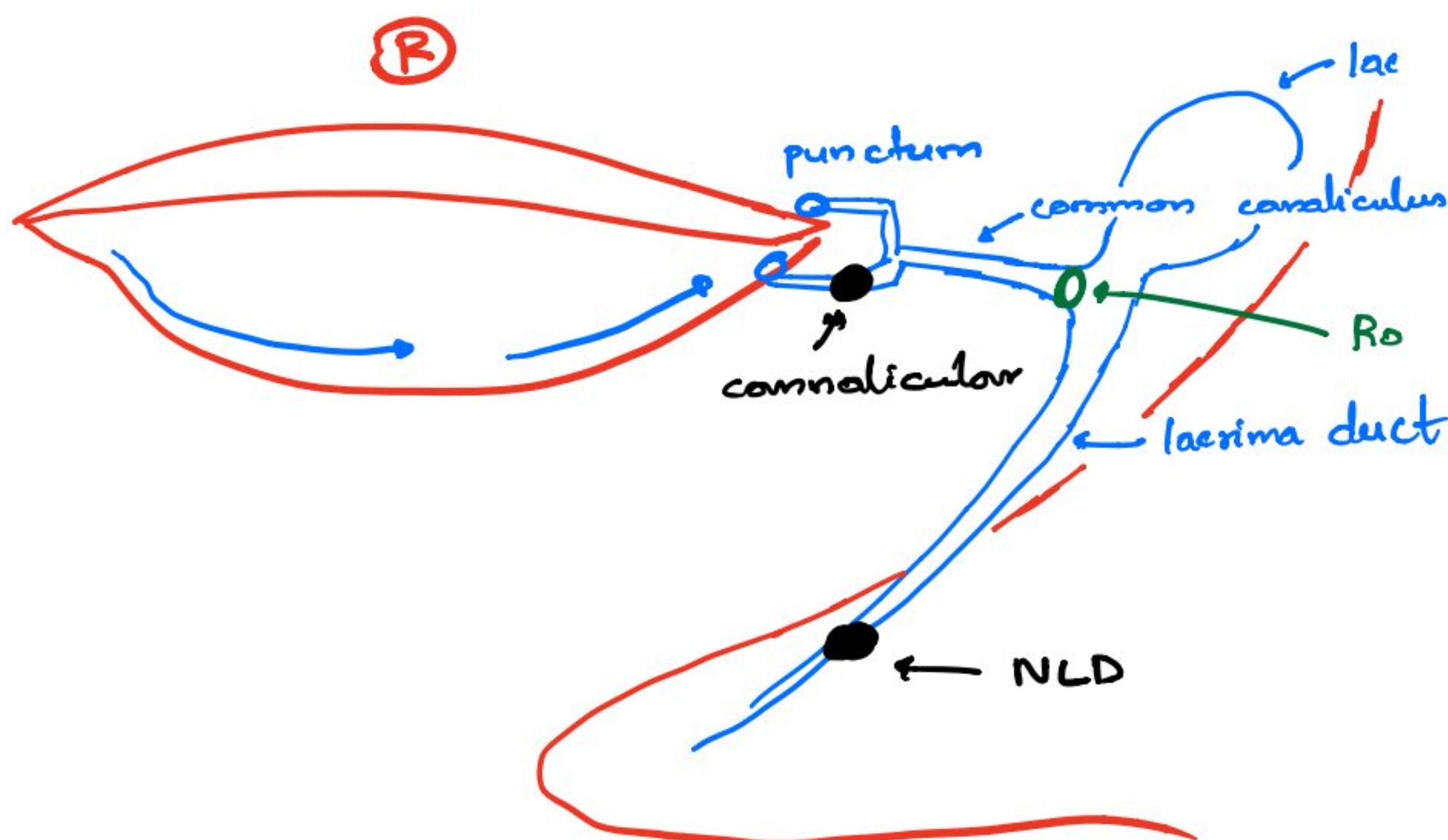
Epiphora

↑ LG activity or obstruction canaliculi Lacrimal pump failure

Inx: JONES dye test

On Bowman Probing:

If probe introduced in puncta reaches medial wall of orbit causing hard stop in patent canaliculi.



NLDO

- No epiphora
- Regurgitation -ve
- Bowman probing → hard stop
- Canalicular obstruction → soft stop.

DACROCYSTITIS



Rx ∴ of NLDO

↓
Congenital

↓
< 9 month = Giggler } over the NLD area
massage

9m - 4yr = Bowmans Probe → mechanically break the obstruction

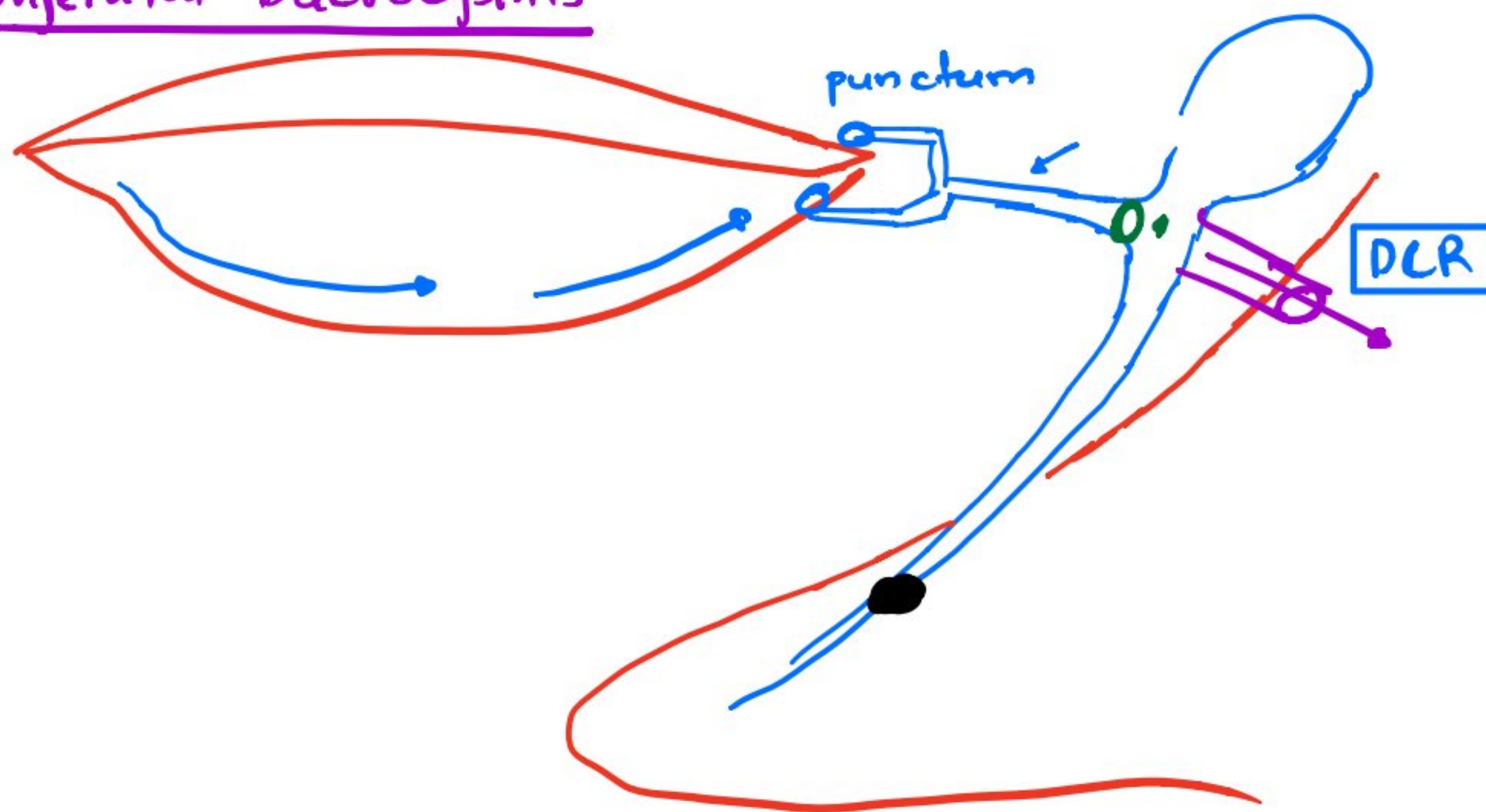
> 4yr = DCR & Dacryocystorhinostomy

Development of
orbital bones completed

Maxillary & lacrimal bone broken in DCR operation
⊕ Nasal concha.

New opening → middle meatus.

Congenital Dacryocystitis



Q: In congenital dacryocystitis
the block is at the ____
a) Lacrimal canaliculi
b) Nasolacrimal duct
c) Punctum
d) None of the above

Indications of DCR

- i) Age < 4yr
- ii) A/c dacryocystitis → Rx = oral AB → DCR
- iii) Lacrimal sac is fibrosed → DCT
Dacryocystectomy
- iv) Lacrimal sac malignancy
- v) Atropine Rhinitis

ACCESSORY LACRIMAL GLANDS

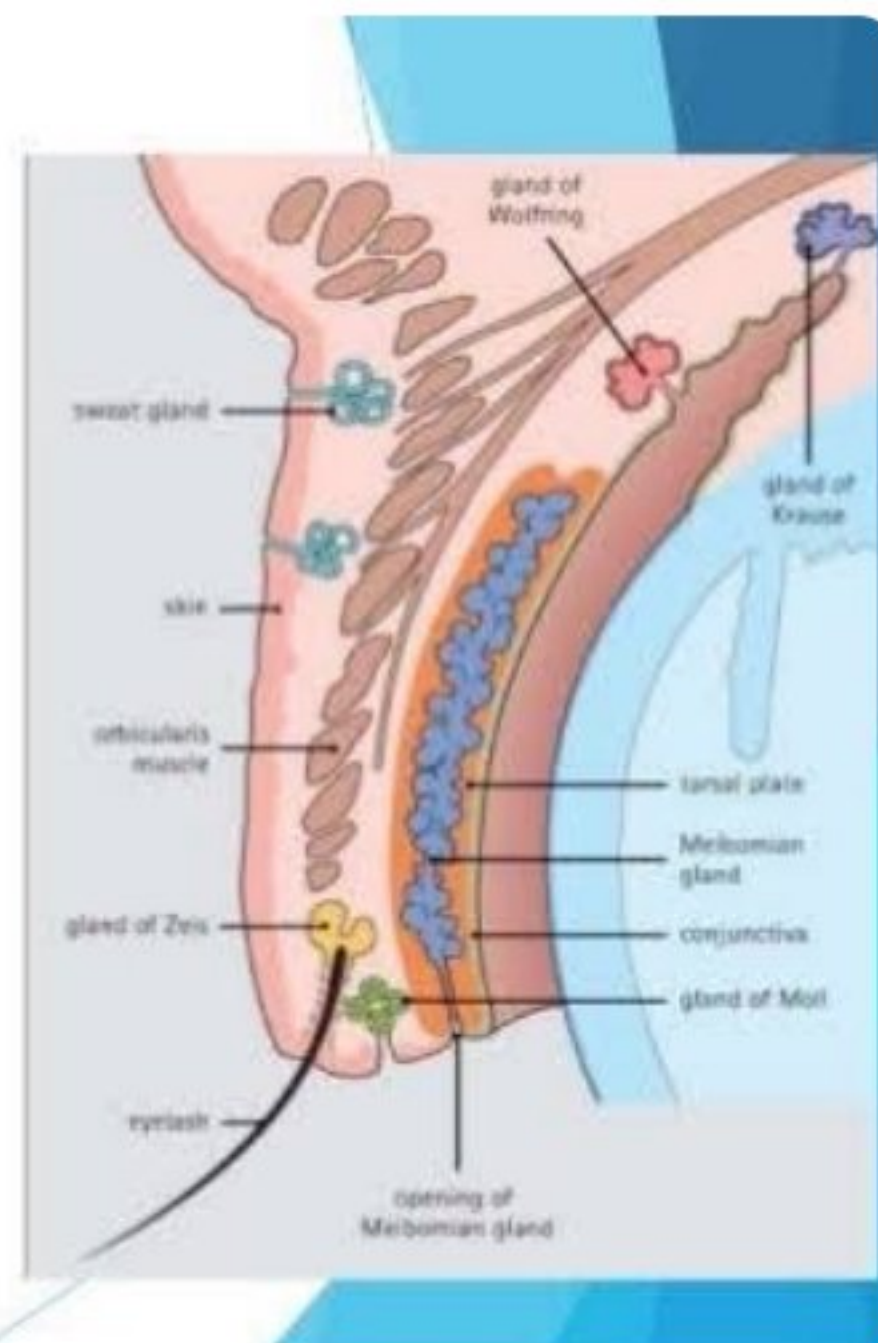
1. Glands of Krause

- lying beneath the palpebral conjunctiva between fornix and edge of tarsus.
- about 42 in the upper fornix and 6-8 in the lower fornix.

2. Glands of Wolfring

- present near the upper border of the superior tarsal plate and the lower border of inferior tarsus.

All lacrimal glands – serous acini similar to salivary glands



24 hr secretion of } by acc. glands
tear

DC's in adults:

A/C

- Painful medial
- Canthal swelling
 τ regergetation test +ve
- org: Staph. aureus
- Rx: Systemic Antibiotics
 & NSAIDs

C/C

- Chronic Painless
- Medial Canthal swelling τ RT $\oplus$

TOC = DCR

org: Staphylococcus Aureus.

Sequale: Lacrimal fistula

Mx of L. sac malignancy:

- Bloody tears
- Rx: DCT = Dacryocystectomy

LG Tumours

- (MC) LG Tumour: Pleomorphic adenoma (NEET'19)
- (MC) LG malignancy: Adenocystic Ca.
- LG malignancy τ worst prognosis: Mucoepidermoid Ca.

(MC)	CHILDREN	ADULTS
1) U/L proptosis	Orbital Cellulitis	TED
2) B/L proptosis	Orbital mets from neuroblastoma	TED
3) Intra-orbital tumour	Dermoid cyst	Cavernous Hemangioma
4) Malignant intraorbital tumour	Orbital mets from neuroblastoma	Orbital mets ♂ = Ca. Prostate ♀ = Ca. Breast smoker = Ca. Lung.
5) 1 ^o malignant intraorbital	Rhabdomyosarcoma	Diffuse large Bcell Lymphoma

Proptosis / DKA / Black eschar



Mucormycosis

- Rx:
- IV Amphotericin B
 - Debridement
 - Exenteration $\{$ Remove all orbital structures $\}$

- (MCC) of blindness → Cataract
- (MCC) of preventable blindness → Vit A deficiency
- (MC) infectious cause of blindness → Trachoma

Tear film break up time

- Tested on mucin layer for adequate mucin content
- TBUT = < 10sec → dry eye

Test for dry eyes

Test ::

- i) Schirmer test
- ii) Tear break up time → Mucus
- iii) Phenol red test
- iv) Tear lysozyme
- v) Tear osmolarity
- vi) Staining of cornea - conjunctiva

→ 2% Fluorescein
→ 1% Rose Bengal
→ 1% Lissamine

SCHIRMER TEST :: req anaesthesia

getting wet > 15mm → (N)
< 5mm → severe dryness

• Schirmer Test I = Total production
(Basal + Reflex)

• Schirmer Test II = Basal Production
[local anaesthesia]

fnx of Main L.G {Reflex production} = STI - STII



5min

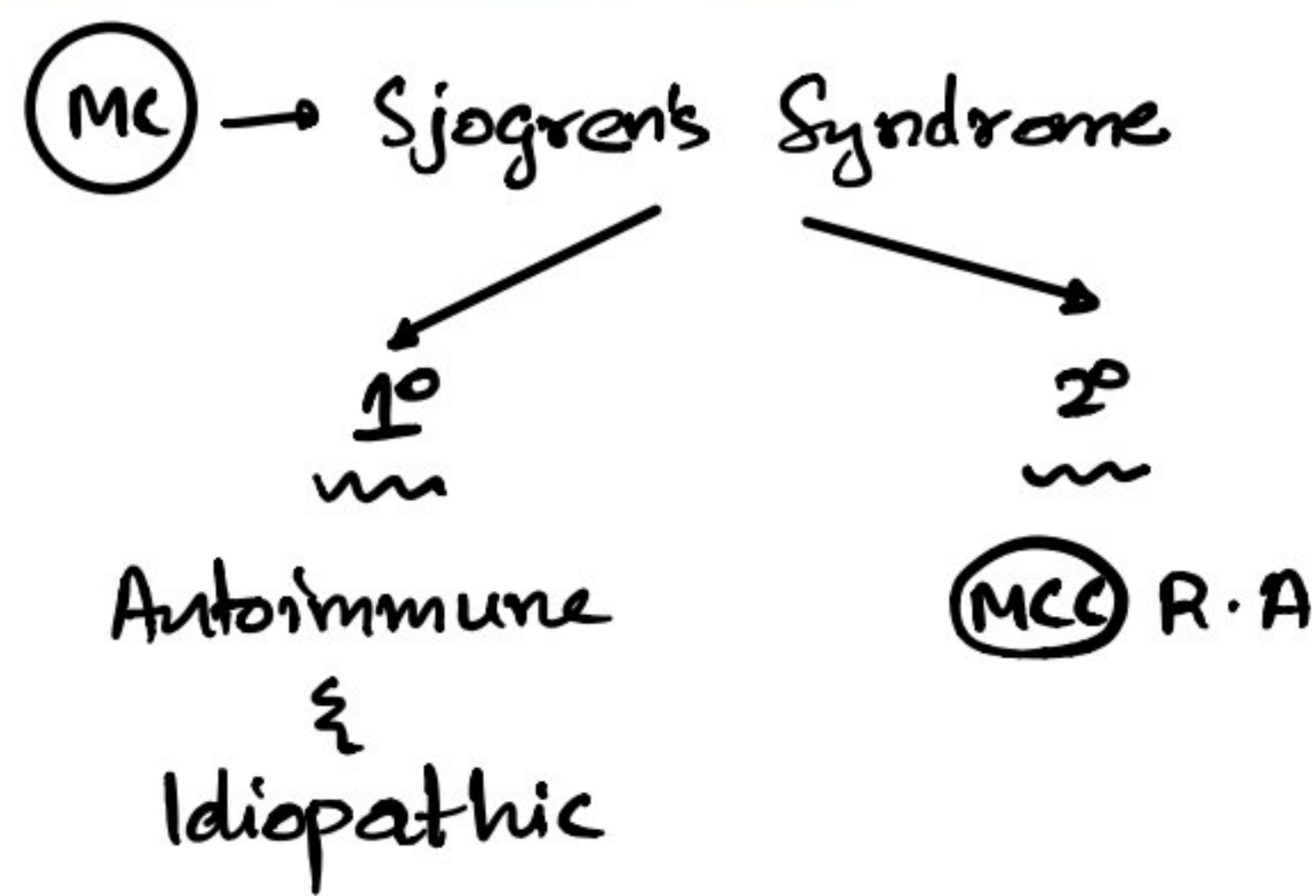
Severe Dryness :: Parenchymatous Xerosis

- Eg::
- i) Keratoconjunctiva sicca → MCC is Rheumatoid Arthritis
 - ii) Steven Johnson Syndrome
 - iii) Chemical injuries →
 - Alkali more dangerous than acid
 - can penetrate the eye
 - $\text{Ca(OH)}_2 \Rightarrow$ (MC) Alkali
 - $\text{H}_2\text{SO}_4 \Rightarrow$ (MC) Acid.
 - $\text{NH}_3 \Rightarrow$ Most dangerous
 - HF → most dangerous acid

Rx:: i) Wash your eye

- ii) Slit lamp examination
- iii) Topical lubricants
- iv) Topical steroids [only in initial stages]
- v) Topical citrate / ascorbate
- vi) Limbal stem cell transplantation → Keratoprosthetics

Keratoconjunctivitis Sicca { deficiency of aqueous layer }



Deficiency of tears → Hyperosmolality of tears

MIKULIKZ SYNDROME :-

- B/L C/L enlargement of lacrimal gland
salivary gland

AIMS :-

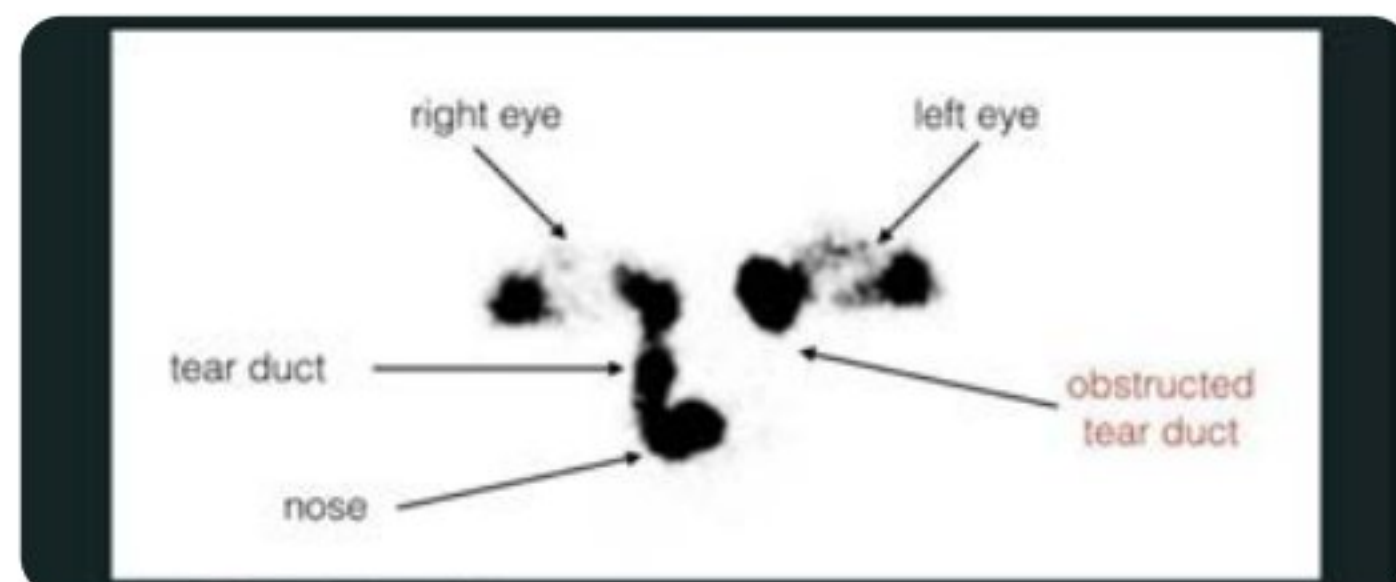
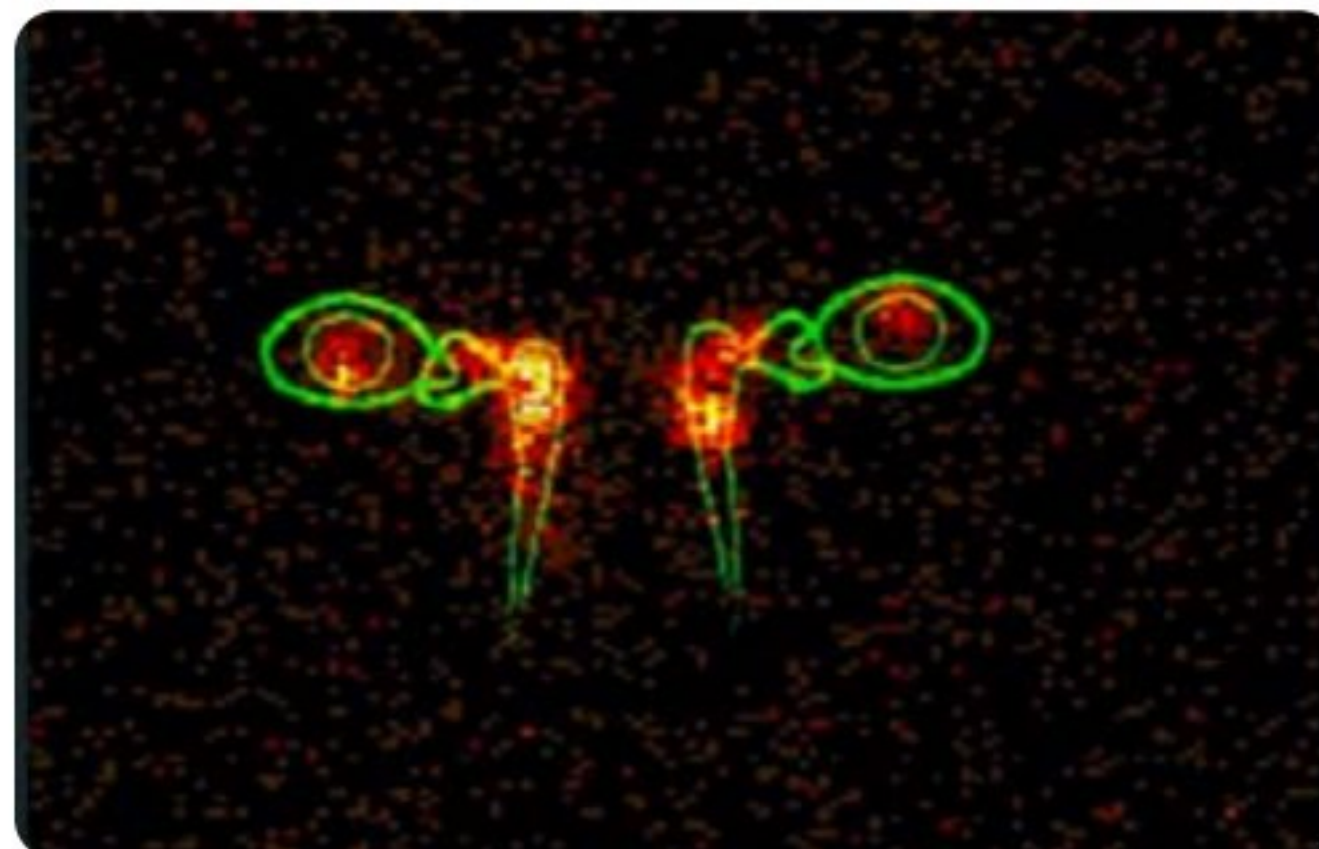
Phenol Red test :-

Measures the volume of tears based on the length of the colour change in the thread

LACRIMAL SCINTIGRAPHY :-

NEET PG Identify the test represented by the following image :-

- a) Regurgitation test
- b) Dacrycystography
- c) Jones dye test
- d) Lacrimal scintigraphy.



Dacryoscintigraphy

Dacryoscintigraphy → It is the confirmatory test for lacrimal pump failure

Q A 60yr old man presents with watering from right eye since 1yr. Syringing revealed a patent drainage system. Rest of the ocular examination was normal. A provisional diagnosis of lacrimal pump failure was made. Confirmation of the diagnosis is done by

a) Dacryocystography

b) Dacryoscintigraphy

c) Syringing

d) Canaliculus irrigation test

Test for evaluation of watering eye.

- 1) Regurgitation test
- 2) Syringing
- 3) Fluorescein dye disapp. test
- 4) **Jone dye test**

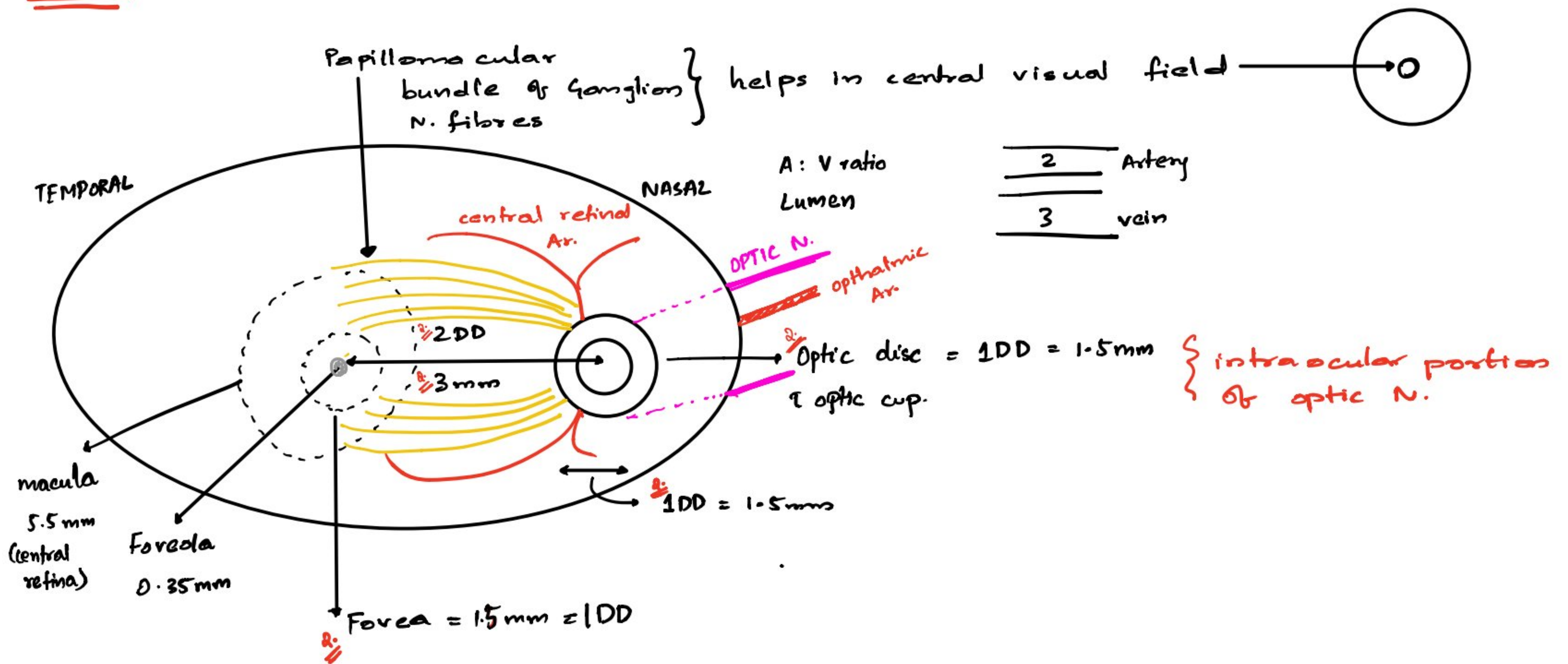
1st test

Diff. epiphora
d/t to obstruction
of lacrimal passage
or d/t the 1st hypersecretion
of tears

2nd test / Irrigation test

Epiphora d/t functional
obstruction → lacrimal pump failure
or a physical obstruction

RETINA



Note:

Retinal thickness is variable in the center. The retina is thinnest at the foveal floor and thickest at foveal rim.

At ora serrata, the retina is the thinnest of all

• Fovea in (R) side = (R) eye
in (L) side = (L) eye

• Retina is colourless / transparent
Red colour d/t choroid

Foveola

is the most sensitive point of Retina.

• Cones are maximum in density

Red Green

Absent at foveola {
① Rods
② Blue cones
③ Ganglion cell layer.



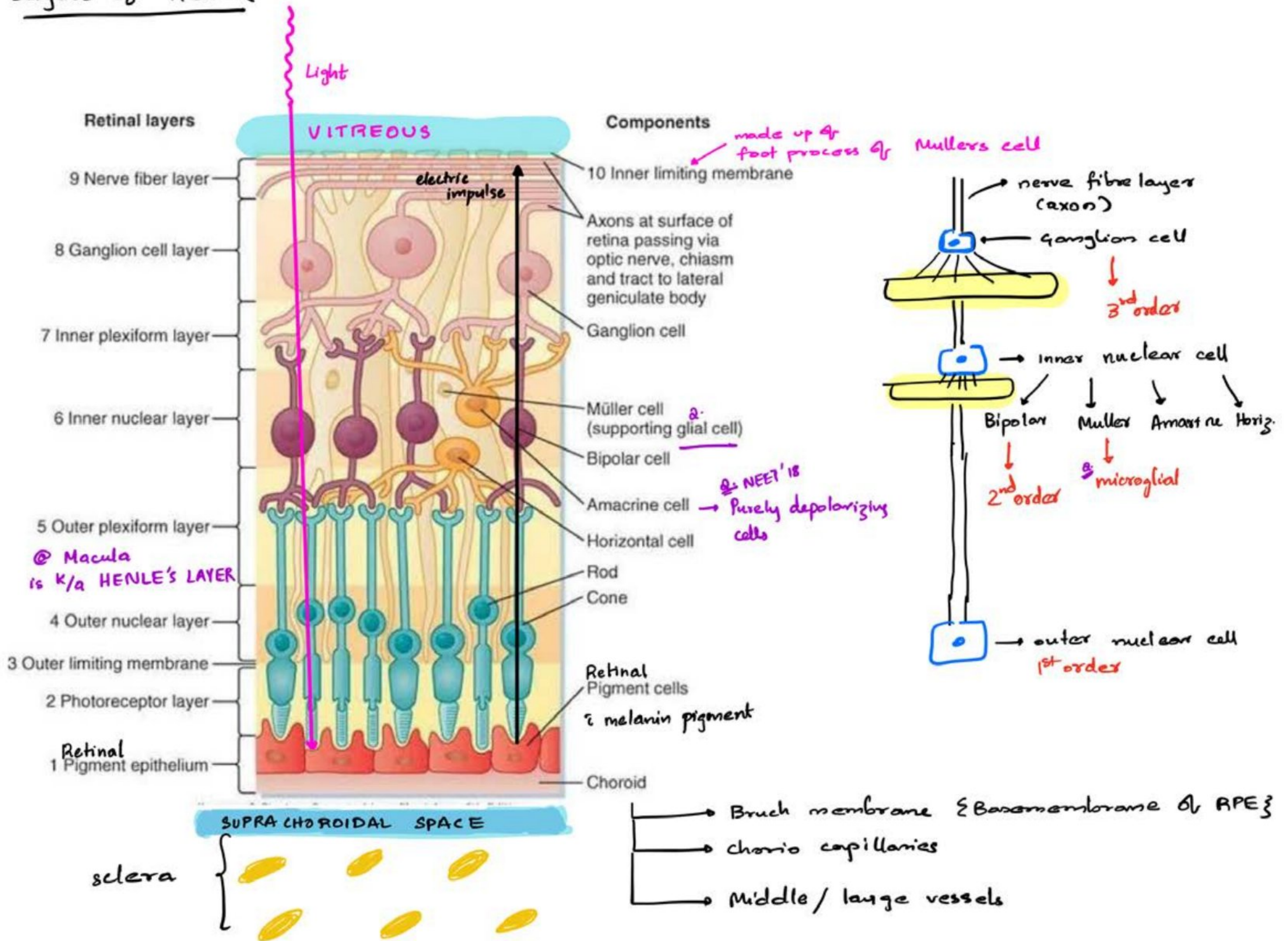
In fundoscopy, veins are bigger than artery.

Posterior pole of Retina $\Rightarrow$ Macula + Optic disc

Optic disc $\rightarrow$ colour = pink,
Thickness = 1mm

Thinnest position of Retina = ORA SERATA
{ peripheral Retina }

Layers of Retina:



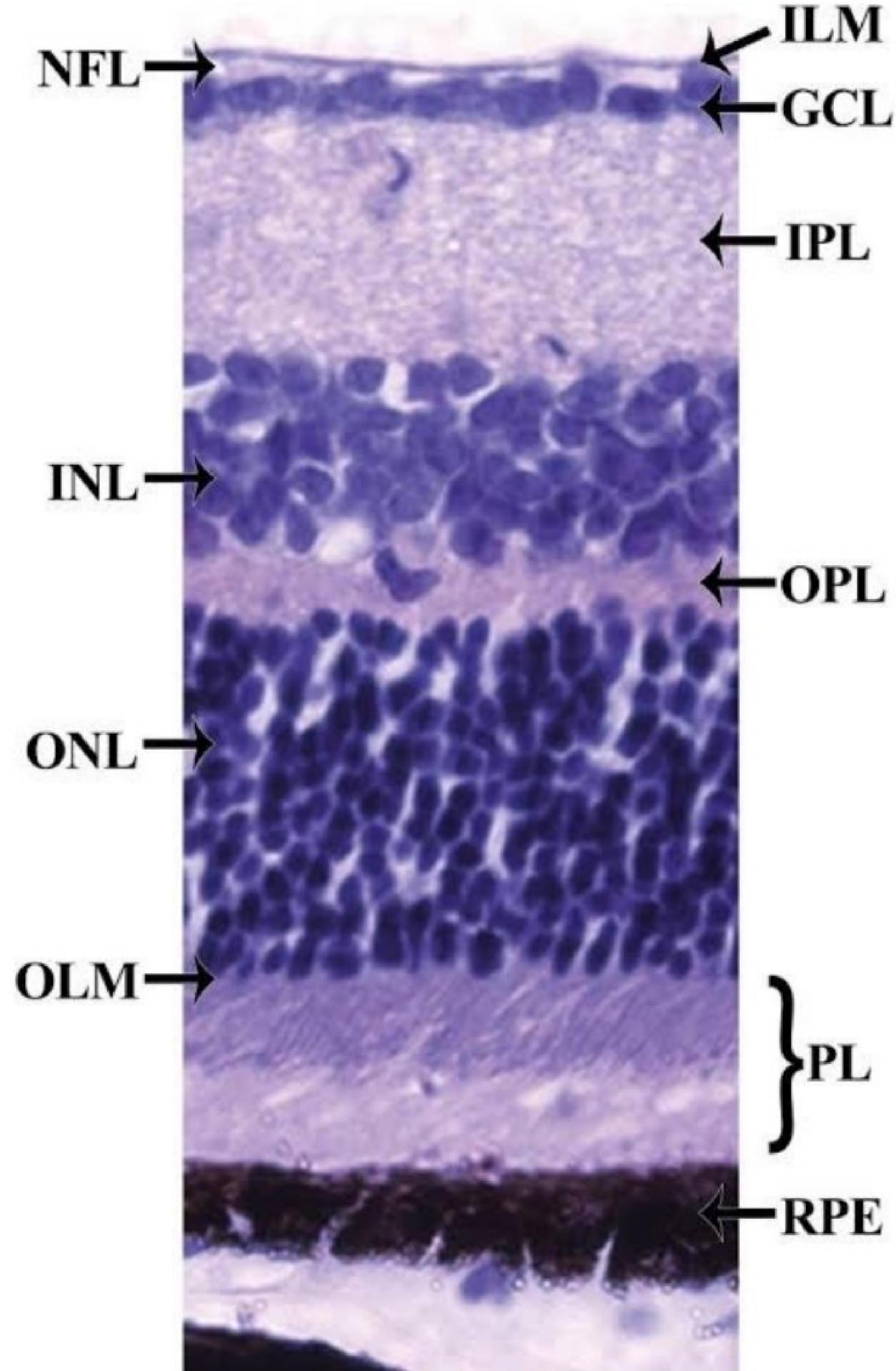
• Rods & cones are the most radiosensitive layer of Retina.

• 5 layers are present in foveola $\rightarrow$

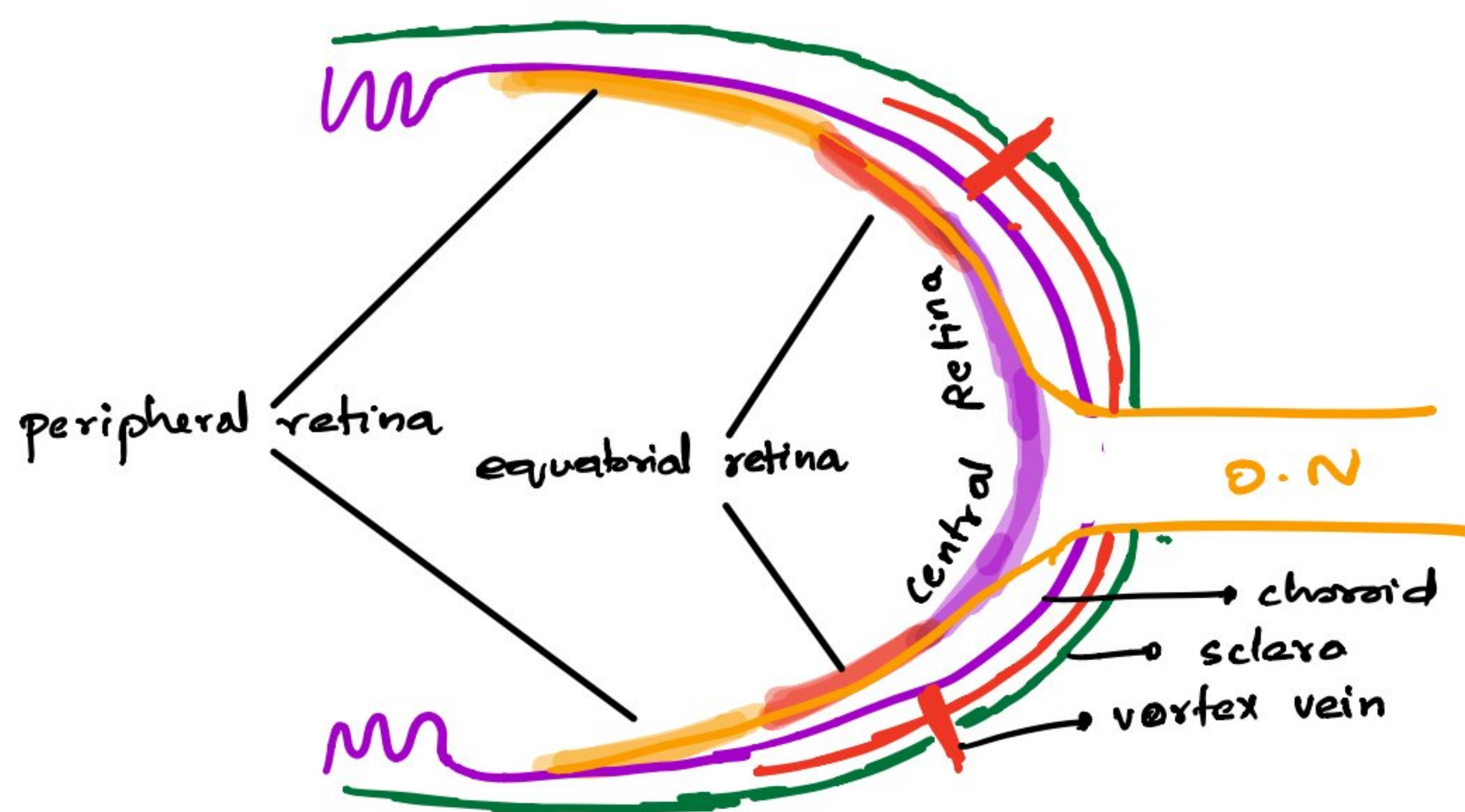
- ILM
- OLM
- cones
- ON
- RPE

• 6 layer are present in fovea $\rightarrow$ ④ OPL

• Rest all 10 layer



Retinal Zones

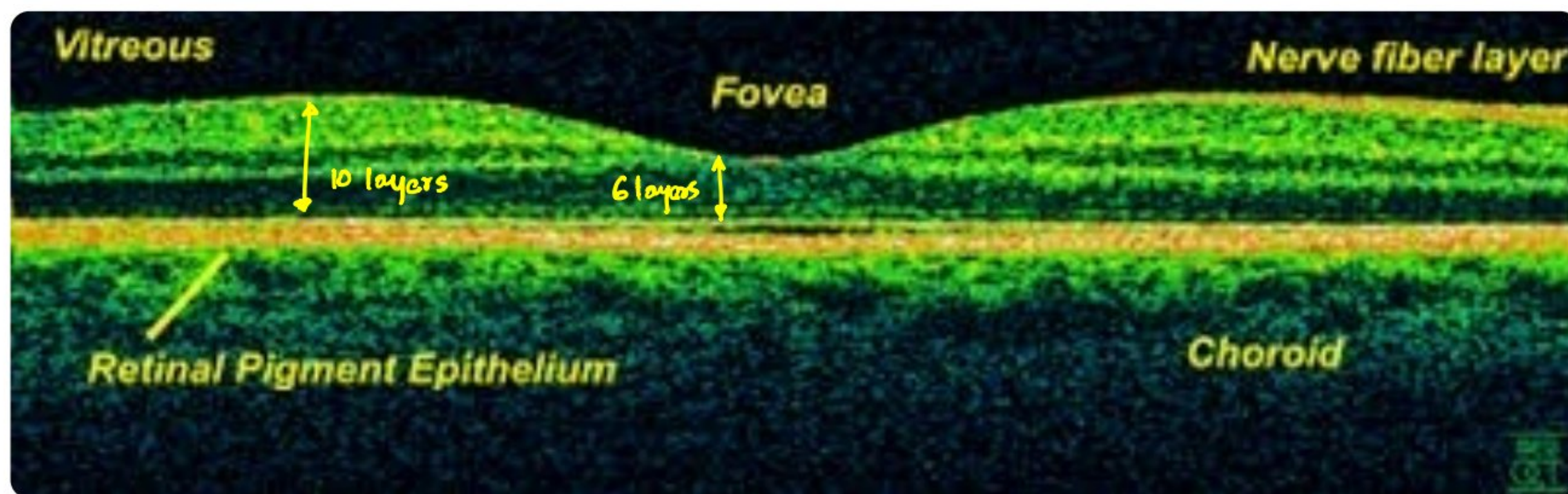


VIEW

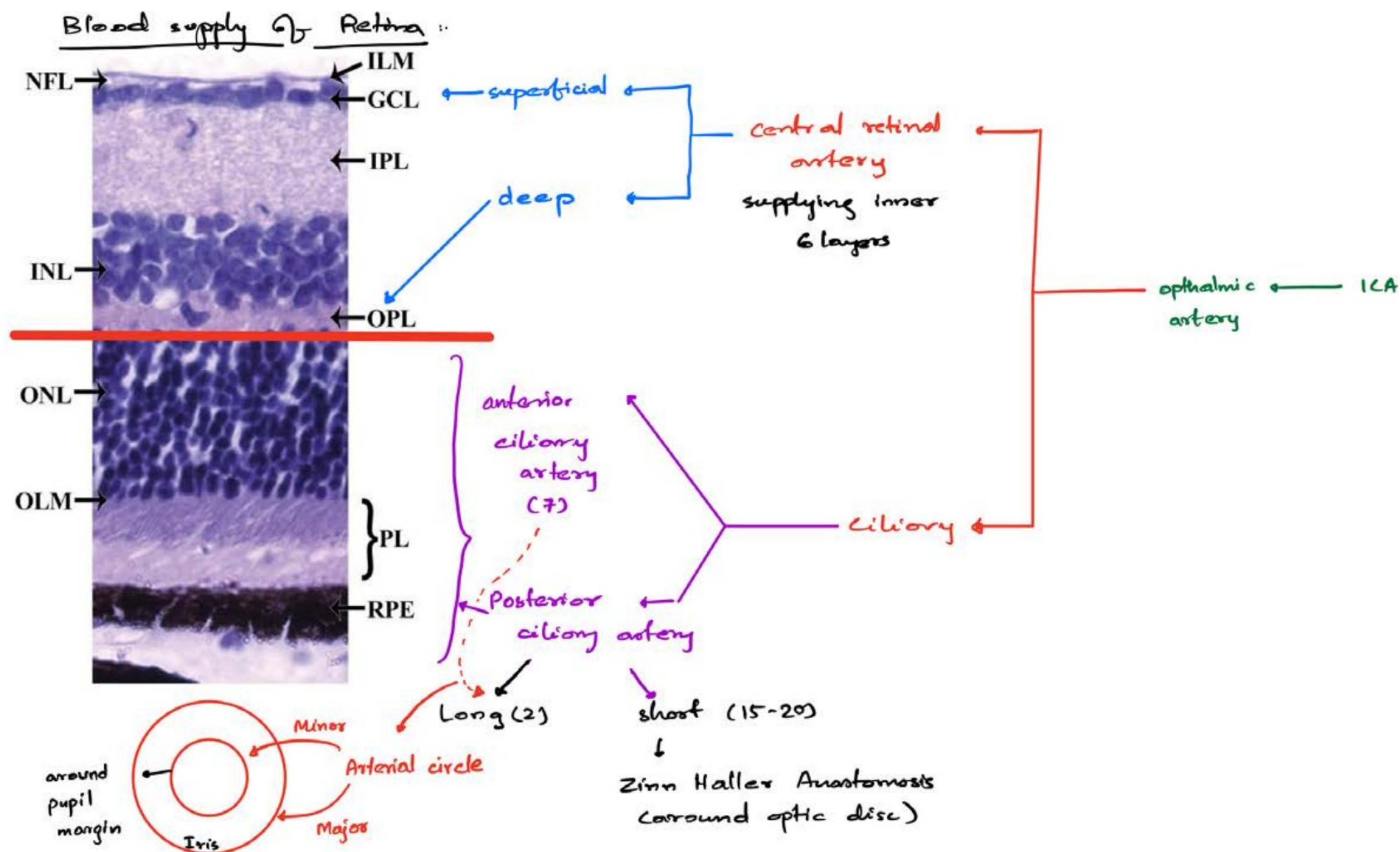
- 1, Direct ophthalmoscope → upto equatorial retinae
- 2, Indirect ophthalmoscope → Complete Retinae
- 3, Slit lamp +90D → optic Disc & Macula
- 4, Goldmann's 3 mirror → " " "

OCT : Optical Coherens Tomography

Outer plexiform layer → a.k.a Henle layer → Radial configuration of central retina

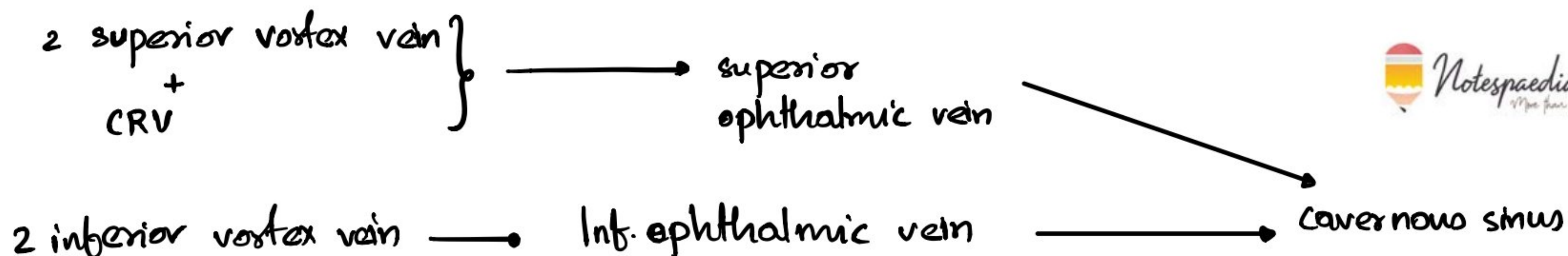


- Ora serata → Thinnest part of retina
- Macula → Thickest part of retina



- Inner blood - retinal barrier → Tight junction of Retinal vascular endothelium
 - ↳ Outer plexus : RNFL
 - ↳ Deep plexus : Inner nuclear layer
- Outer blood - retinal barrier → Retinal pigment epithelium { Tight junction b/w RPE }
- Anterior ciliary artery → 2 artery for each rectus except lateral rectus.

Venous drainage

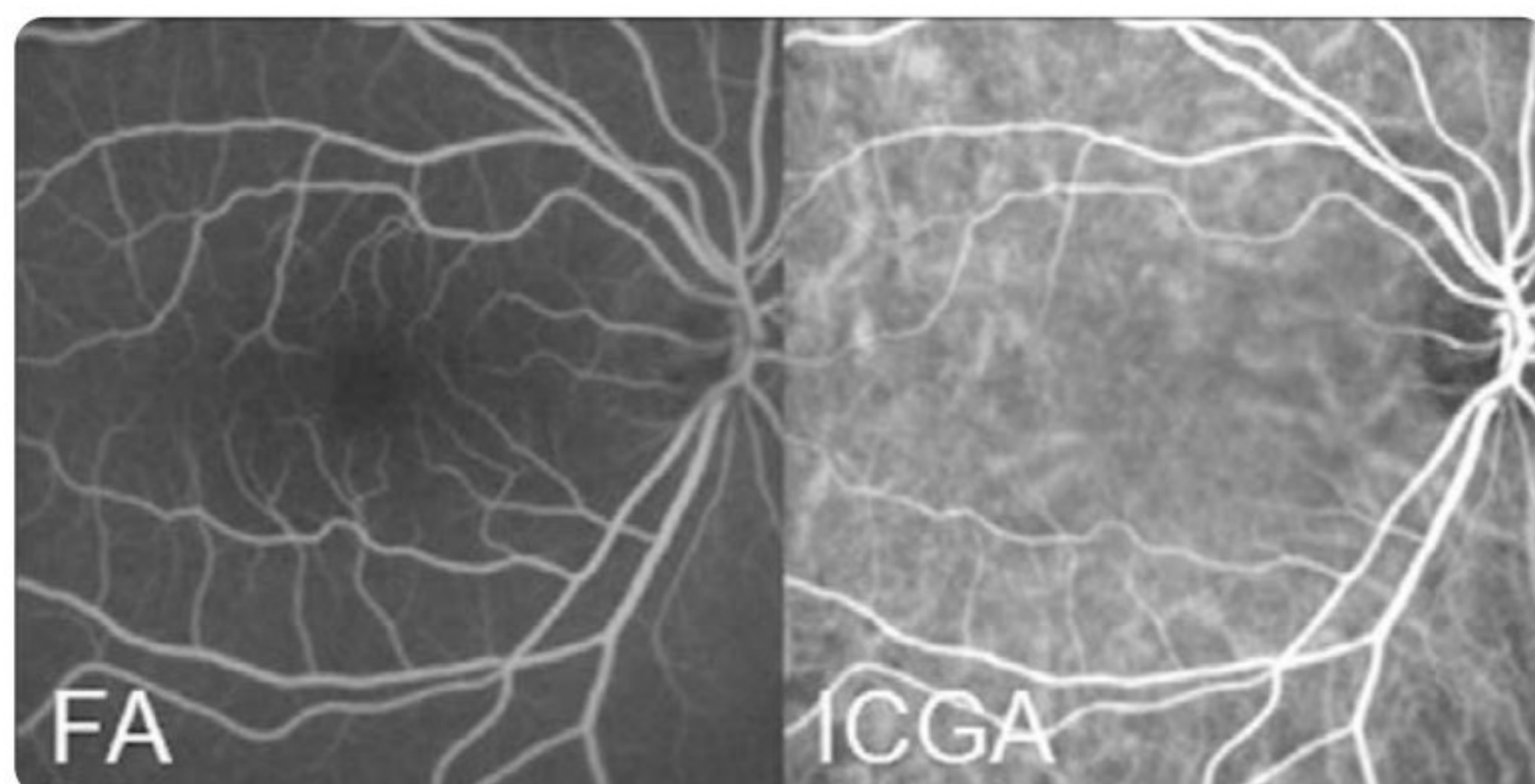


pg 107

- Q: The following are true about optic disc
- a, Forms an absolute scotoma in the eye field T
 - b, Optic disc measures vertically 1.8-4mm & horizontally 1.76mm T
 - c, Approx 1m. nerve fibers are present T
 - d, Supplied by central retinal artery T
 - e, Fovea centralis situated nasally to disc F
(Temporal)

Optic disc → neural retinal rim around optic N. → B.S central retinal artery
 Optic N. → B.S short posterior ciliary artery

ANGIOGRAPHY



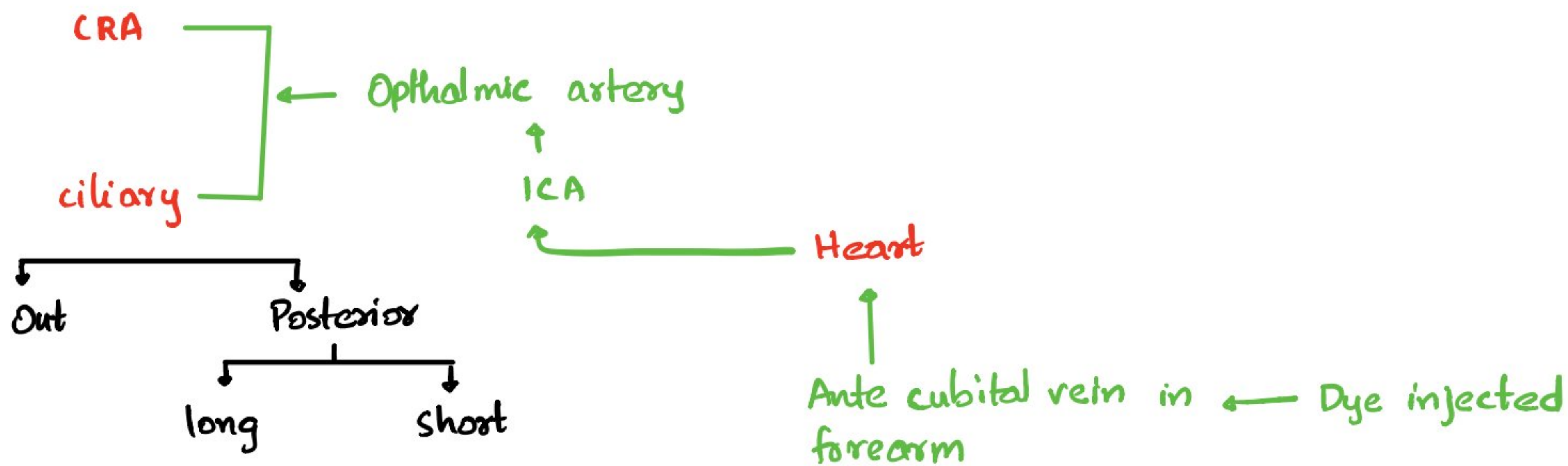
Fovea remains dark d/t melanin pigmented darkly.

Fundus Fluorescein angiography	ICG angiography {indocyanin green}
For retinal circulation	For choroidal circulation
Dye used - sodium fluorescein	Dye used - indocyanin green
80% plasma protein bound and low MW	98% plasma protein bound and high MW
Light of visible spectrum used	Infrared spectrum of light used

These dyes have a property → Fluorescence

incoming light : low wavelength
 outgoing light : High wavelength

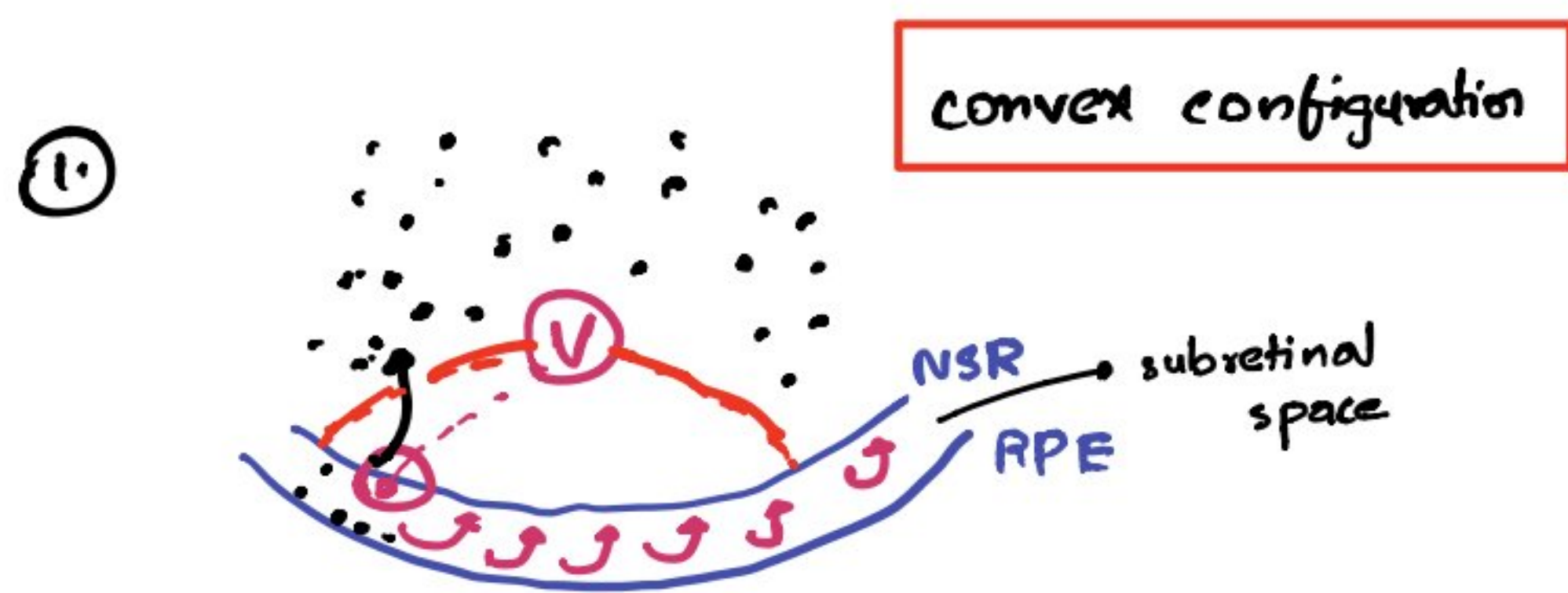
Indication of ICG angiography → occult choroidal neovascular membrane (CNVM)



Retinal detachment: Inner 9 (neurosensory layer) separate from 10th RPE

- Retinal Detachment
- ① **Rhegmatogenous**
 - Retina tear / hole + liquid vitreous
 - convex configuration

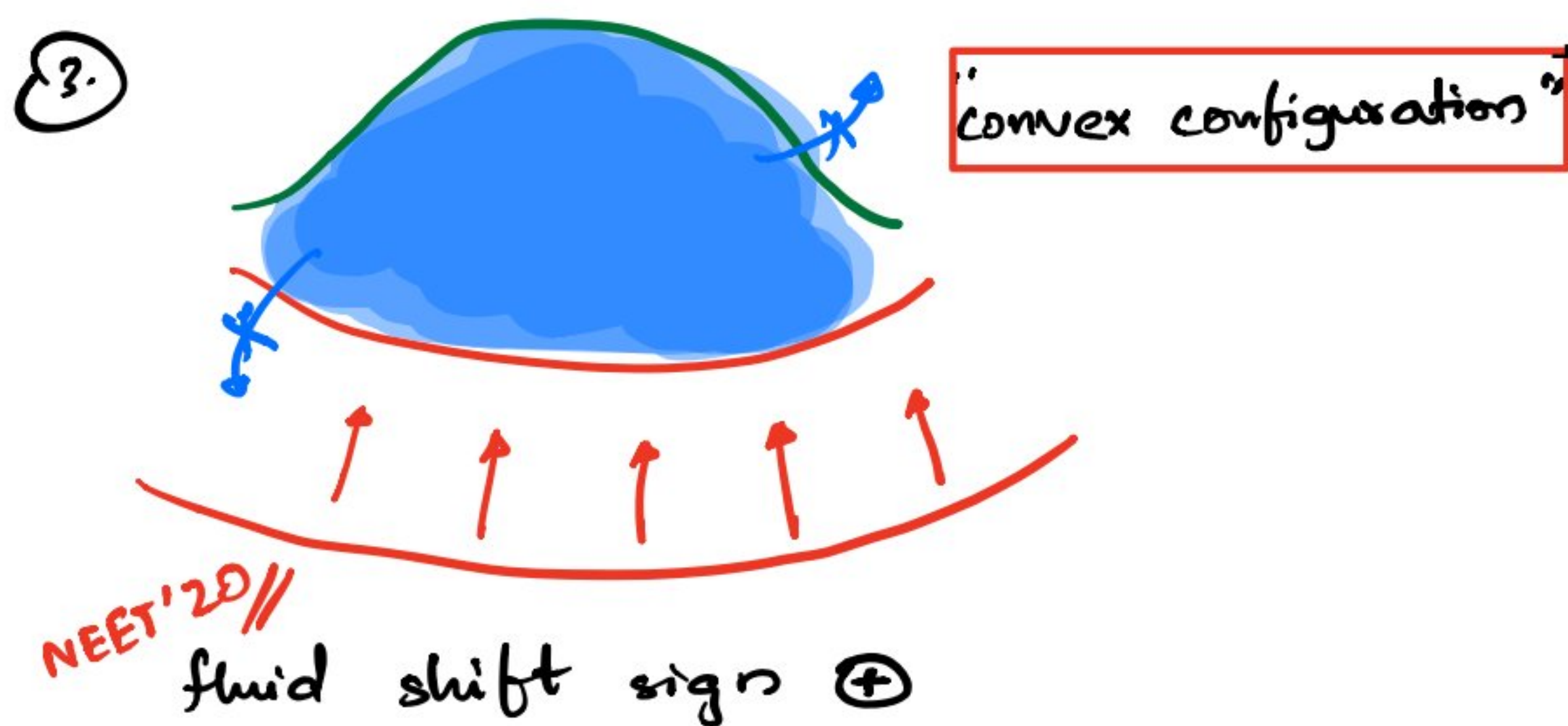
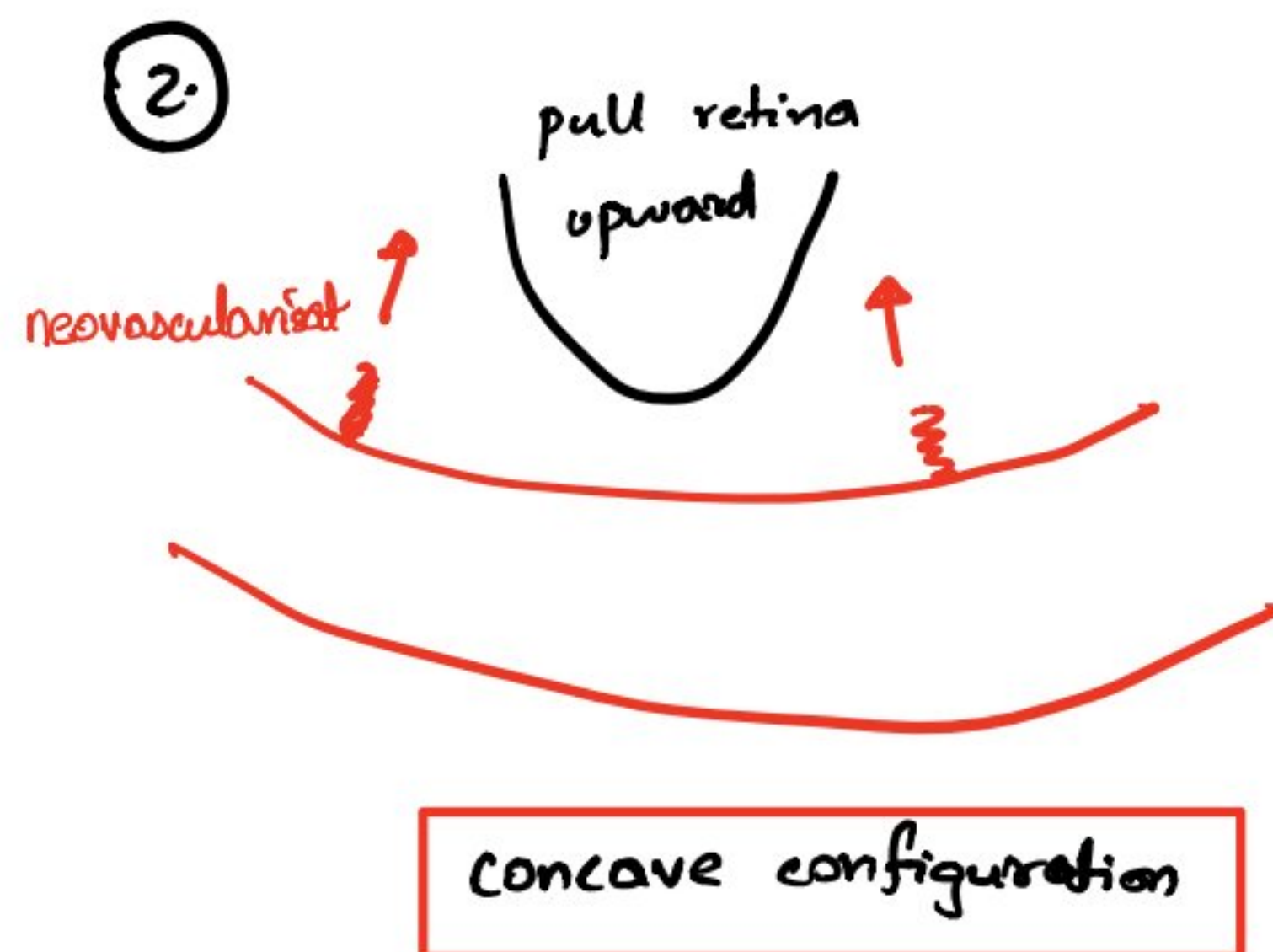
} PVD
Horse shoe shape
 - ② **Tractional**
 - New vessels on retina
 - ③ **Exudative**
 - ↑ fluid from choroid
 - Outer barrier break by inflammation → i) Post. uveitis
ii) Post. scleritis
iii) Choroidal tumour
iv) Pregnancy
v) Central serous retinopathy



Q: **Tobacco dust** ⇒ **Schaffer sign**

symptoms :-

- floaters
- ↓
- flashes / photopsia
- ↓
- field defect
- ↓
- sudden painless loss of vision



① RHEGMATOGENOUS R.D

Symptom:

- Sudden painless loss of vision [if Macula is involved]
otherwise: field defect in the detached area.
- Flashes of light (PHOTOPSIA)
- Curtain falling down symptom

Sign:

- Detached retina → corrugated & convex
- Total RD → Greyish white reflex @ pupil
- Shaffer's sign: Tobacco dusting of vitreous

AIIM '14 Water mark: Subretinal demarcation line
Old rhegmatogenous retinal detachment.

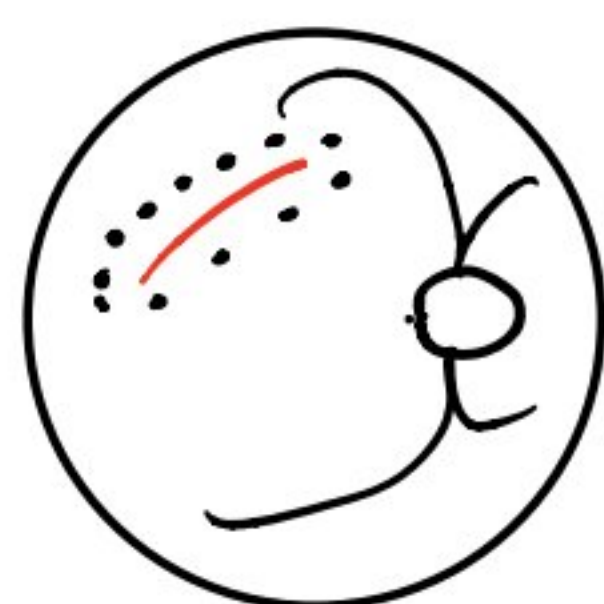
Imx: 1) Indirect Ophthalmoscopy
2) USG.

Causes of RRD:

- 1) Pathological / High myopia
- 2) Lattice Retinal degeneration [Abn thinning of Retina]
- 3) Trauma [penetrating]
- 4) Marfan's Sx.

Mx: Steps → (I): Identify all retinal breaks by thorough IDO
LINCOLN's rule ⇒ to localise the retinal breaks based on configuration and location of RD

(II): Seal all retinal breaks
using: i) Laser "Barrage laser" [scar formation & damping]

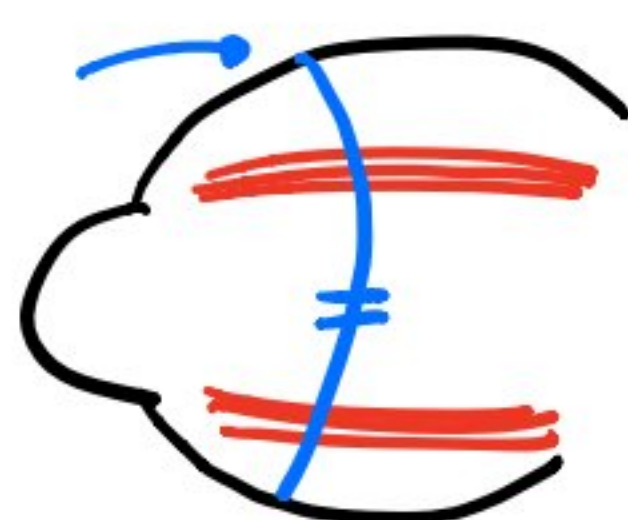


ii) Cryo - only for peripheral breaks

③ Reattach the detached retina

i) External Tamponade → scleral Buckle

scleral
buckle

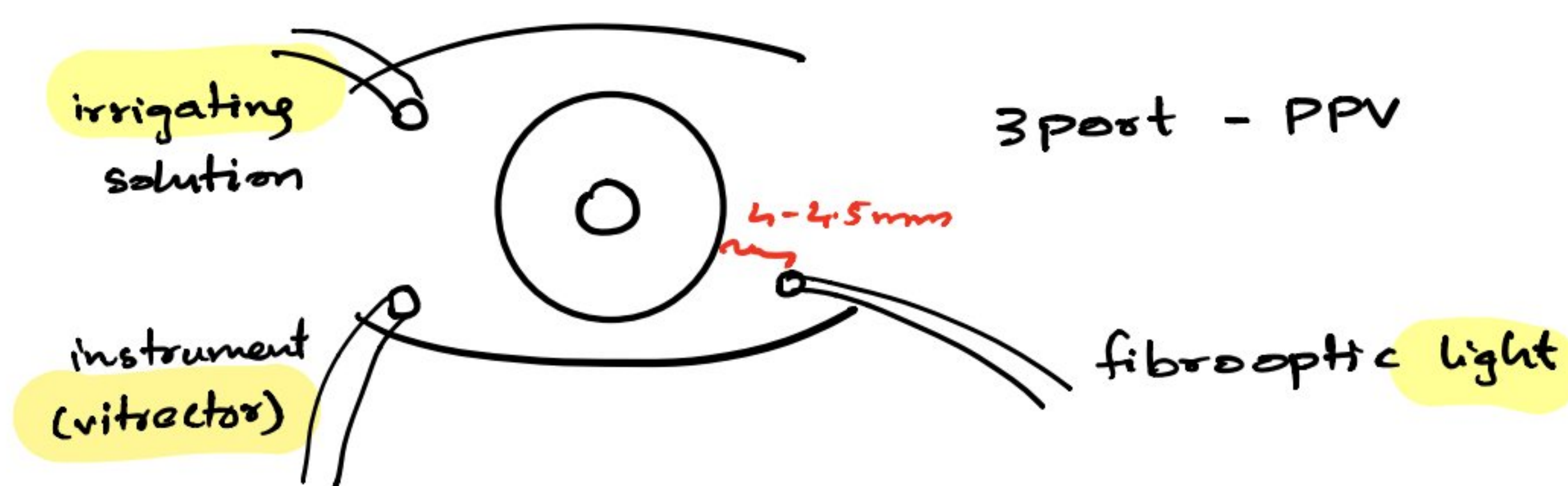


→ Extraocular Sx: pushes RPE towards neurosensory Retina

ii) Internal Tamponade ⇒ vitrectomy

↓
vitreous substitute
& pushes neurosensory retina
toward RPE

Vitrectomy: Par plana = 4mm away from Limbus



size of instrument = 23G

Vitreous substitutes:

i) Gases → i) Non expansile gases

① sterile air

② Nitrous oxide : G.A is C/I d/t 2nd gas effect

ii) Expansile gases:

① SFG & Sulphur hexafluoride ⇒ MC

② C_3F_8 → 0.3ml, largest size reached at 72hr

Avoid Airplane
/ High altitude
travel

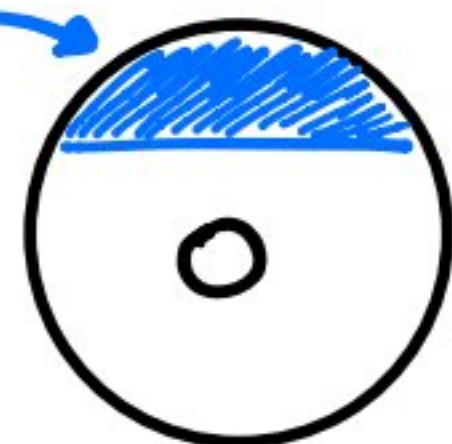
ii) Liquid → i) Silicone Oil

↓
2nd surgery
for removal.

• Inverse hypopyon

• 2° glaucoma

• Ca^{2+} deposition → Band shape keratopathy

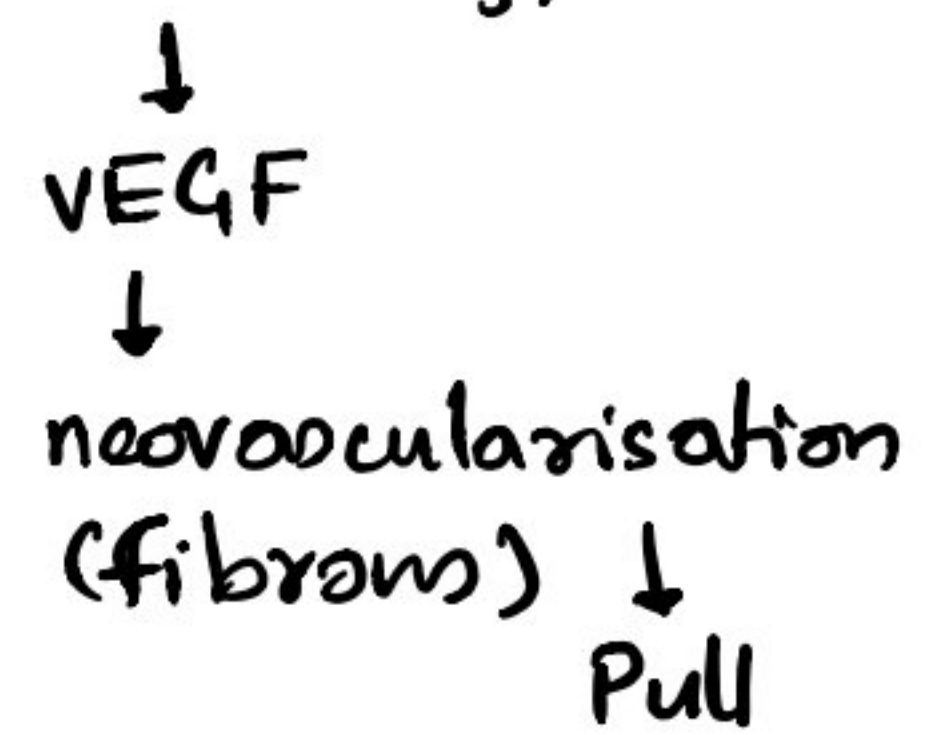


ii) PFCL & Perfluoro carbon liquid

3) Tractional RD → (fibrous) neovascularisation

Pathogenesis :: Retinal hypoxia

- causes
- 1) Penetrating posterior segmental ocular trauma
 - 2) Diabetic retinopathy
 - 3) Sickle cell retinopathy
 - 4) Myopia



Diagnosis :: IDO findings

- fibrous band in vitreous
- neovascularisation
- Epiretinal membrane

Mx :: i) Prep vit. omit VEGF (↓ intra op bleeding)
↓ f/b

ii) Pars plana vitrectomy (remove all fibrous bands)

} simultaneously manage the underlying cause.

↓ f/b

iii) Epiretinal membrane peeling

↓ f/b

iv) PRP: Pan retinal photocoagulation → Endolaser

↓ f/b

v) Vitreous substitute

3) Serous / Exudative R.D :: ^{AIIMS} subretinal fluid d/t ^{Q:} breakdown of outer BRB

Sgnx :: i) IDO findings

- convex smooth detachment of retina
- shifting fluid ⇒ "hallmark"

ii) OCT

iii) USG B Scan

- Causes ::
- i) Toxemia or pregnancy
 - ii) Hypertension
 - iii) Harada's (Vogt Koyanagi)
 - iv) Exudative retinopathy of Coats
 - v) Malignant melanoma of choroid

Mx :: Underlying Rx.

Central serous retinopathy:

• Idiopathic self limited disease of young or middle → aged adult males.

• Type A personality

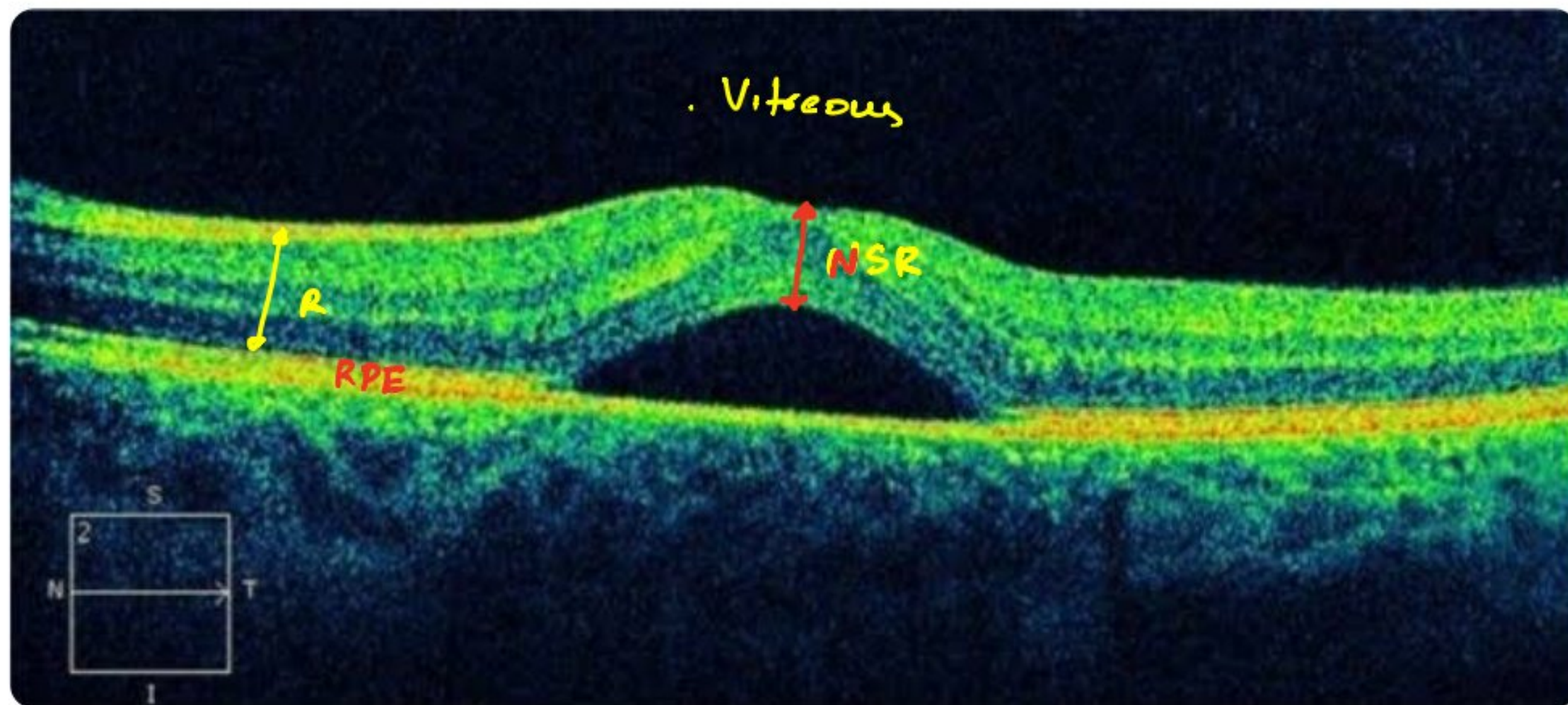
1) Pathogenesis:

Dysfunction of RPE

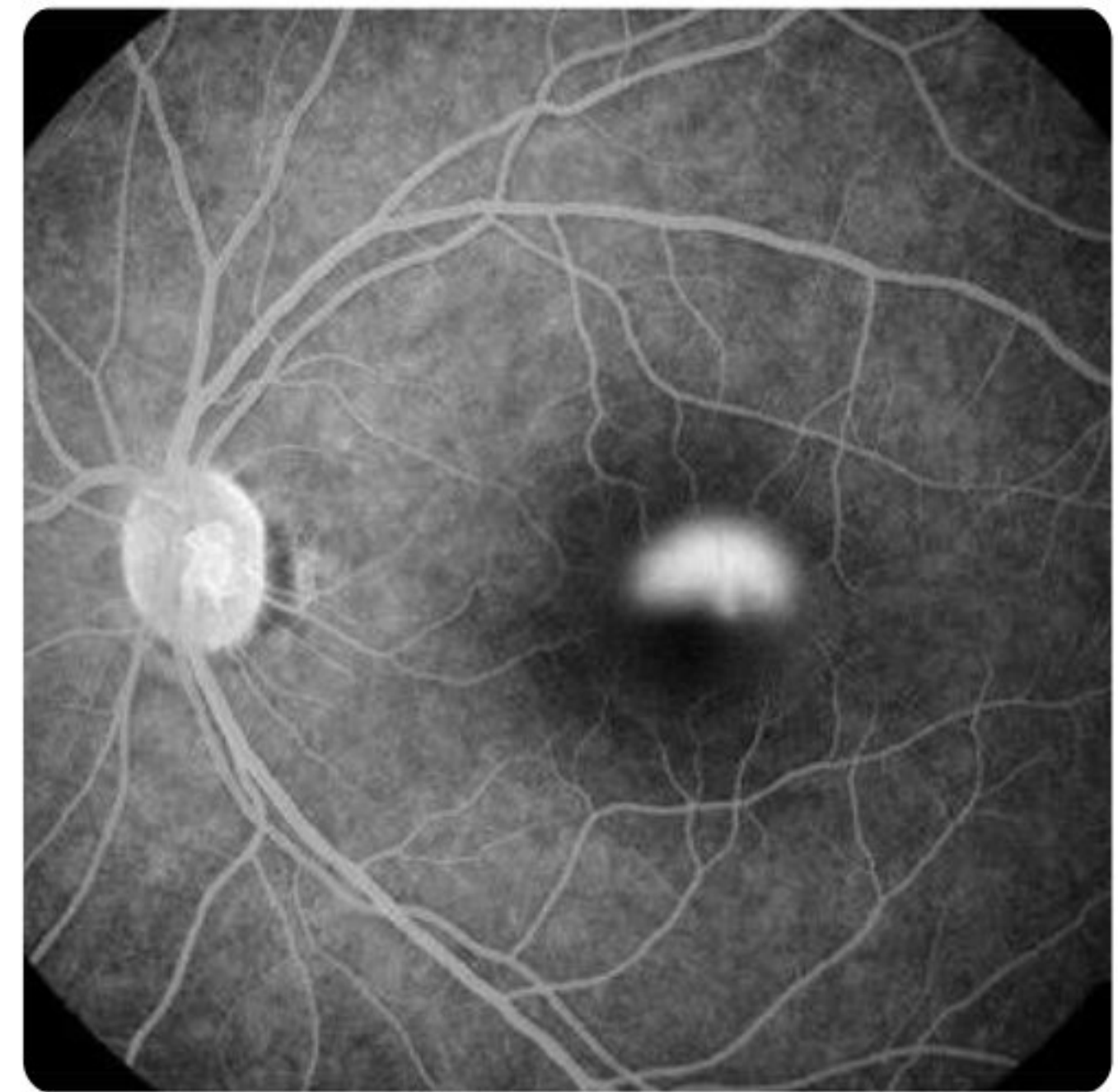
↓
Accumulation of fluid in subretinal space

↓
Local detachment of sensory retina at macula.

2) Risk factor → stress, steroids, smoking.



CSD - OCT



Angiography

C/F: a) sudden onset of blurred vision in one eye

b) With relative central scotoma

c) Micropsia

d) Metamorphopsia

e) V.A is 6/9 → 6/12

} Acq. HYPEROPIA

Q: Ink blot pattern / smoke stack pattern → d/t dye sweeping in to macula.

Rx: Wait for 3 months → Resolve by its own

> 3month

→ Laser

Photodynamic therapy

689nm

Q: Dye used: Verteporphyrin

Q: Identify the condition



a) Foveal detachment

b) macular hole

c) Macular edema

d) Macular Drusen

→ s/i = central serous retinopathy
• sudden loss of vision

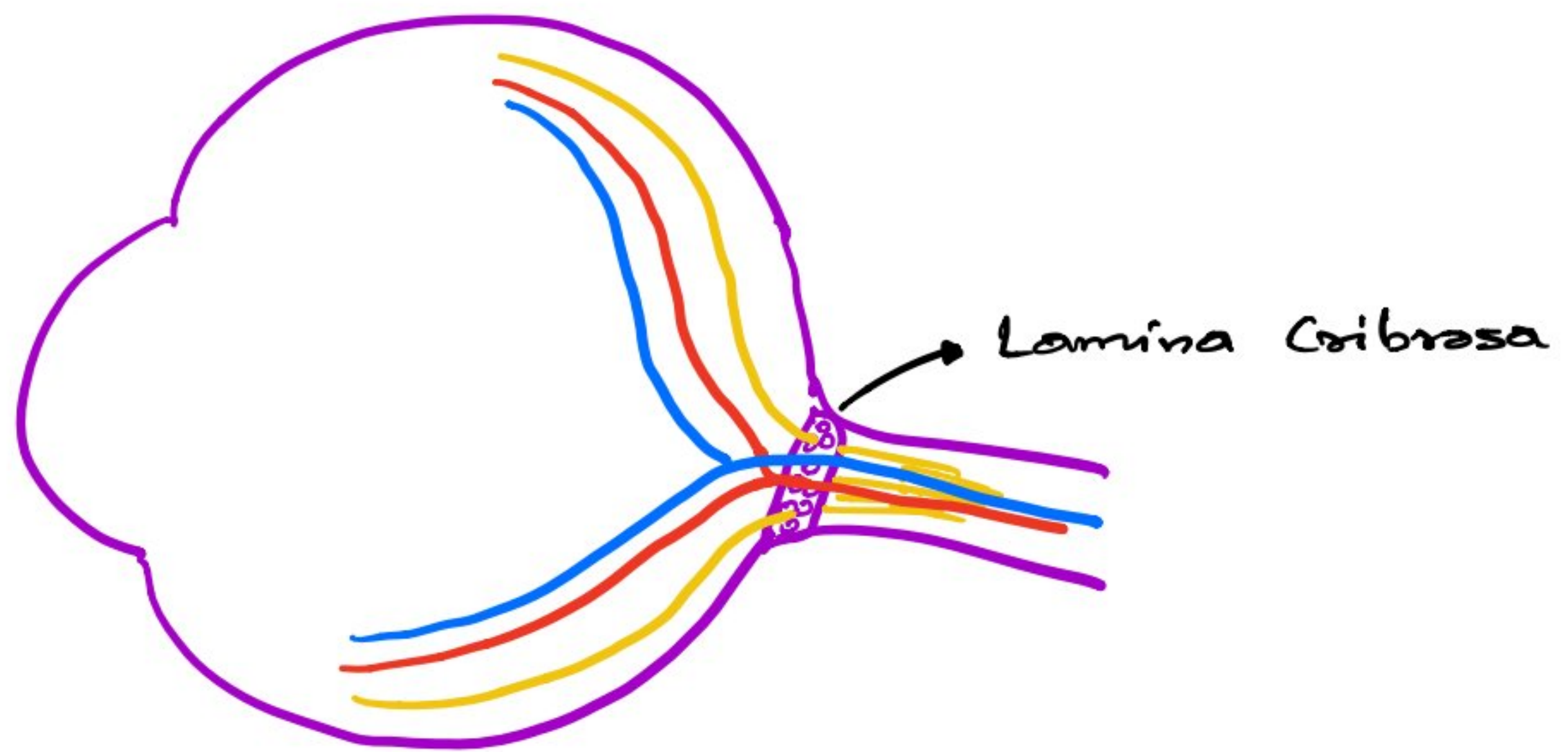
Q. A 10 years old boy after watching the sun during eclipse in the apparent twilight developed persistent negative after image of the sun. He was taken to an ophthalmic surgeon who examined his fundus and found it.

④. Most probable diagnosis

- Ⓐ Malingering
- Ⓑ Photophthalmia
- Ⓒ ☒ Photo retinitis
- Ⓓ Light induced maculopathy

Retinal Vascular Occlusion:

Site of occlusion: Lamina cribrosa → part of sclera @ optic disc where central retinal vessels and optic nerve pierce it to enter Retina



Retina Vein Occlusion:

BRVO: superotemporal retina → only one quadrant is involved

CRVO

site: optic disc ⇒ Lamina Cribrosa

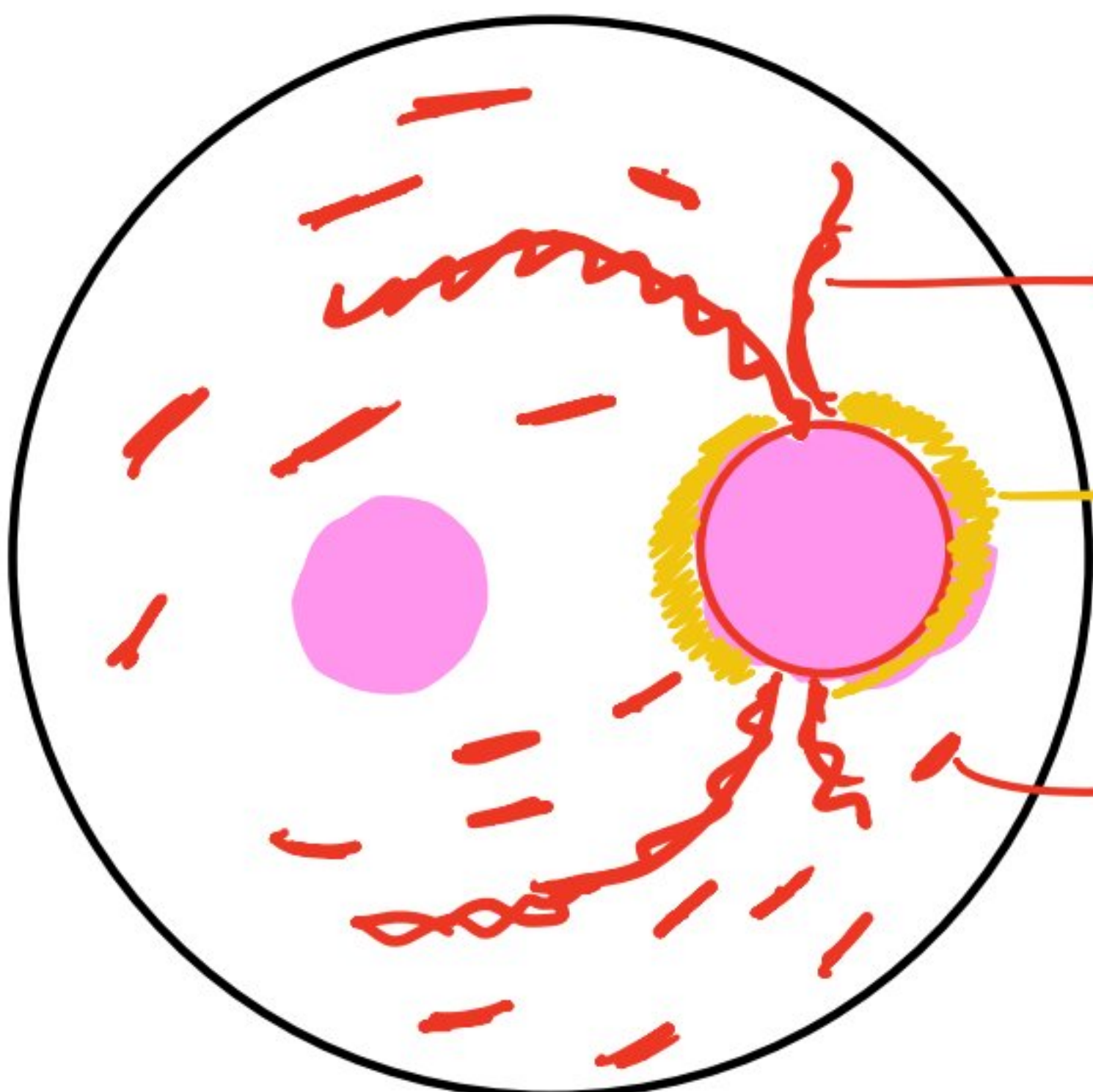
↓
Ⓐ Splashed tomato appearance of CRVO



BRVO

Risk factor

- i) Old age → most important
- ii) HTN
- iii) Atherosclerosis (MC)
- iv) Hyperlipidemia
- v) Diabetes 2nd (MC)
- vi) Smoking and obesity
- vii) Raised IOP.
- viii) Hypercoagulable d/o
- ix) Hyper homocystenemia

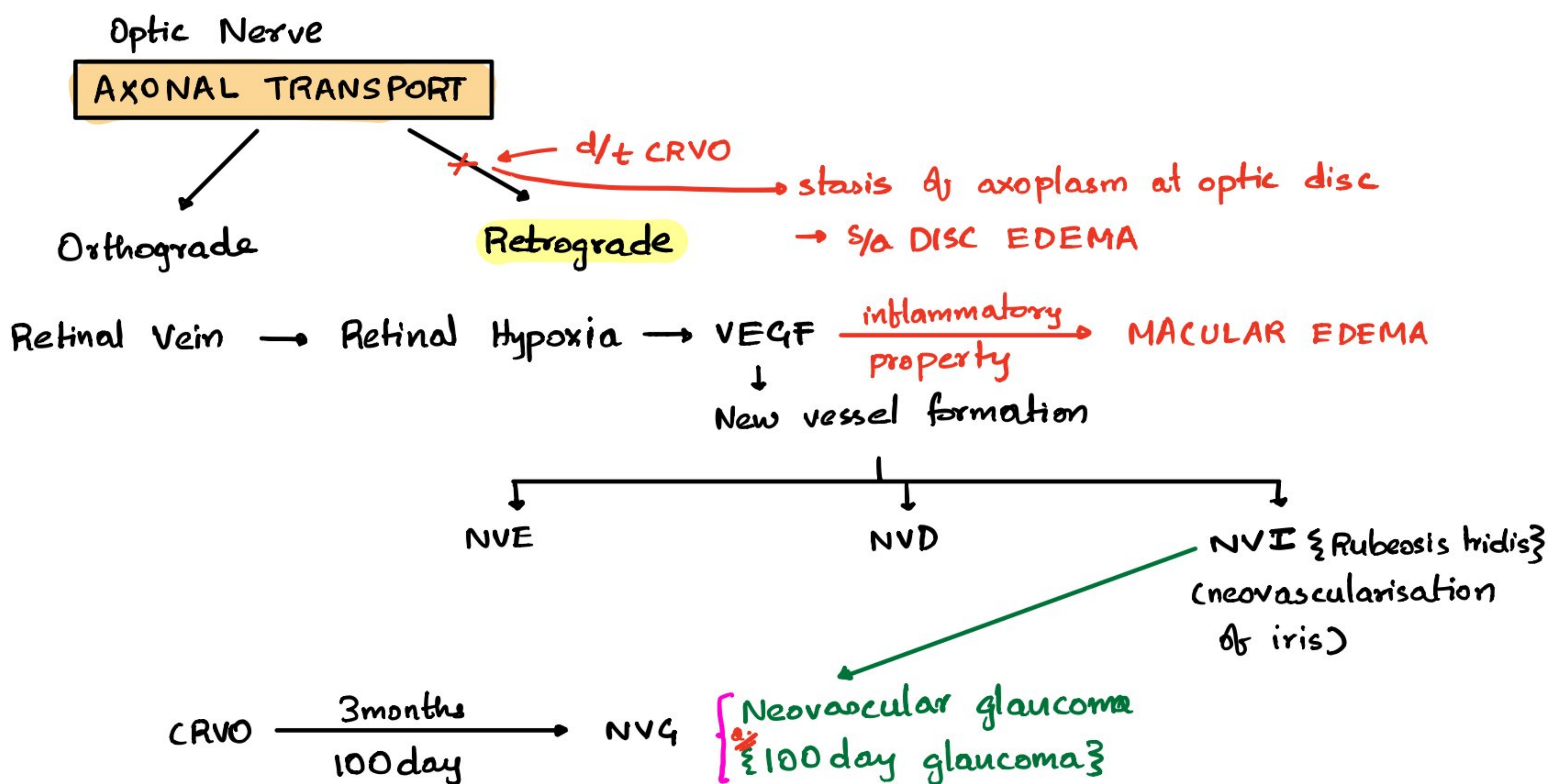


Dilated & tortuous Retinal Vein.

Disc Edema: Axoplasmic Stasis & No real edema?

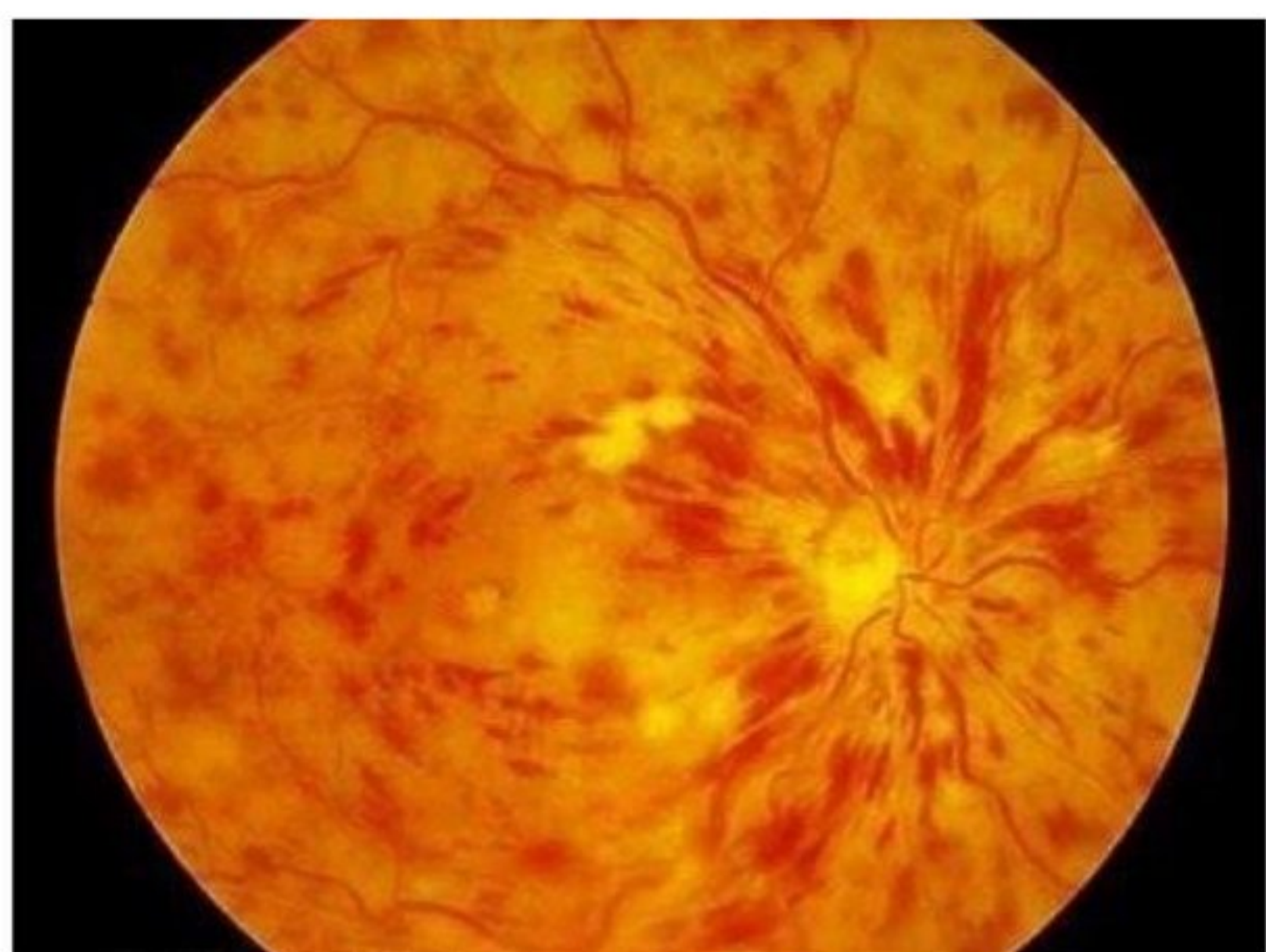
FLAME SHAPED / SPLINTER

Retinal H'mage @ RNFL d/t superf. capillaries



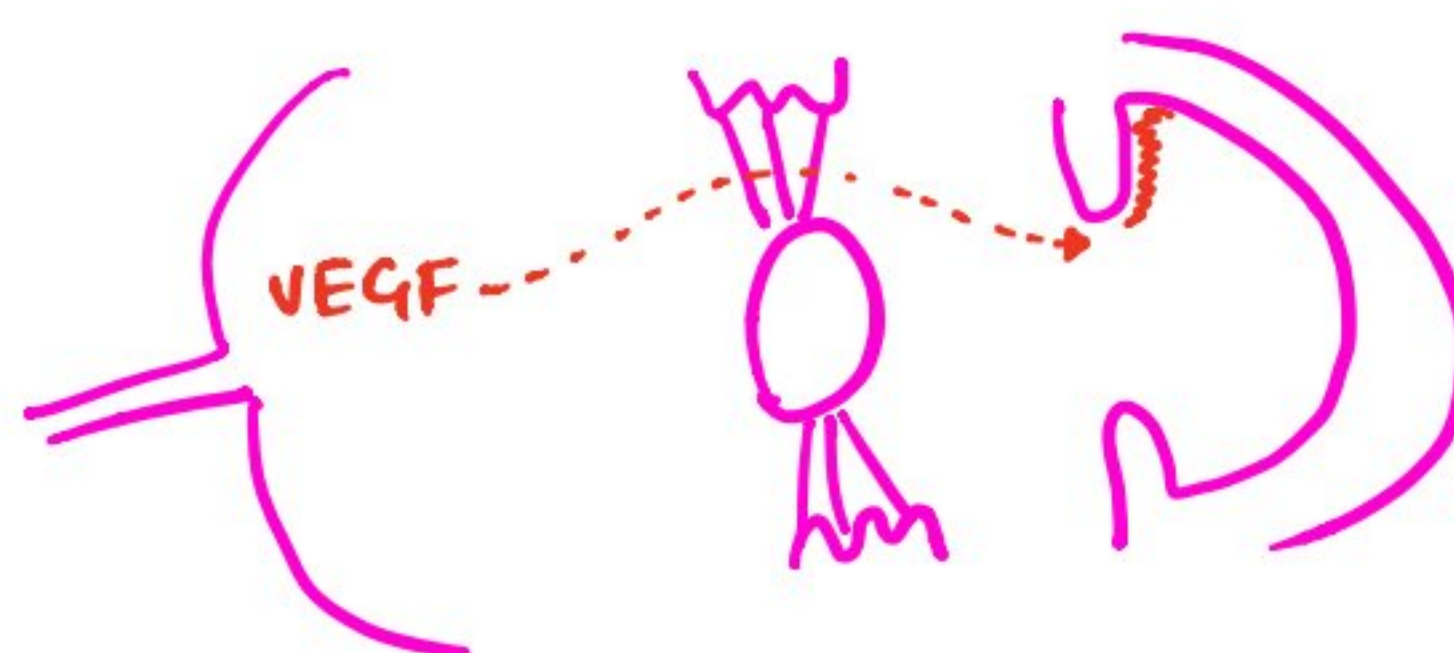
Fundus finding of CRVO:

- 1) Dilated & tortuous retinal vein
 - 2) Retinal H'mage { superficial; flame shaped }
 - 3) Disc Edema
 - 4) Macular Edema → ↓ se in vision in CRVO
- } Splashed tomatoes appearance of fundus



CRVO

VEGF moves to the vitreous and later reach to produce NVI → NVA



2° OAG

2° ACG

{ Neovascular Glaucoma }

Types of CRVO

Non Ischemic

< 10 DD area of retinal H'mage

Ischemic CRVO

- > 10 DD area of retinal H'mage
- RAPD
- Electro Retinogram ⇒ B'wave Absent
- ↓↓↓ vision

Rx → control the underlying cause
CRVO

NVI

- 1° ⇒ Pan retinal Photocoagulation for NVI using Retinal Laser
- 2° ⇒ anti VEGF injection

for Macular Edema : Observation x 3 months

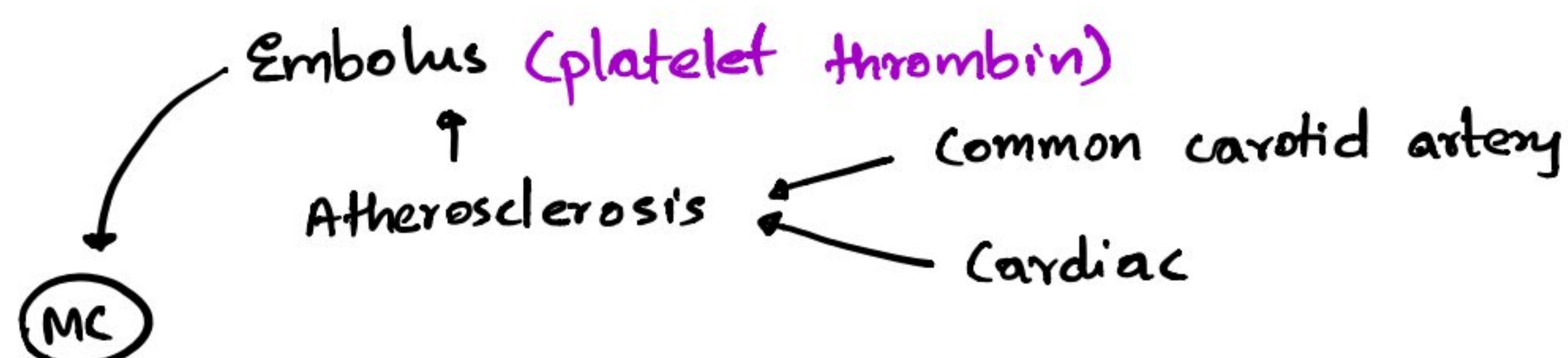
- Grid laser (Diode laser)
- Intravitreal injection of steroid
- " " " anti-VEGF.

Rx of BRVO

- 1° ⇒ only quadrant that is affected } Laser photocoagulation (no photocoagulation)
- 2° ⇒ Anti VEGF injection

Central Retinal artery occlusion : **OPHTHALMIC EMERGENCY**

site :: Lamina Cribrosa



→ Cholesterol & Hellen Horst plaques?

Symptom → **amourosis fugax** ⇒ (D/D) · migraine patient
· giant cell arteritis

sign $\rightarrow$ i) Retinal edema (milky white colour)
L. Ganglion layer edema

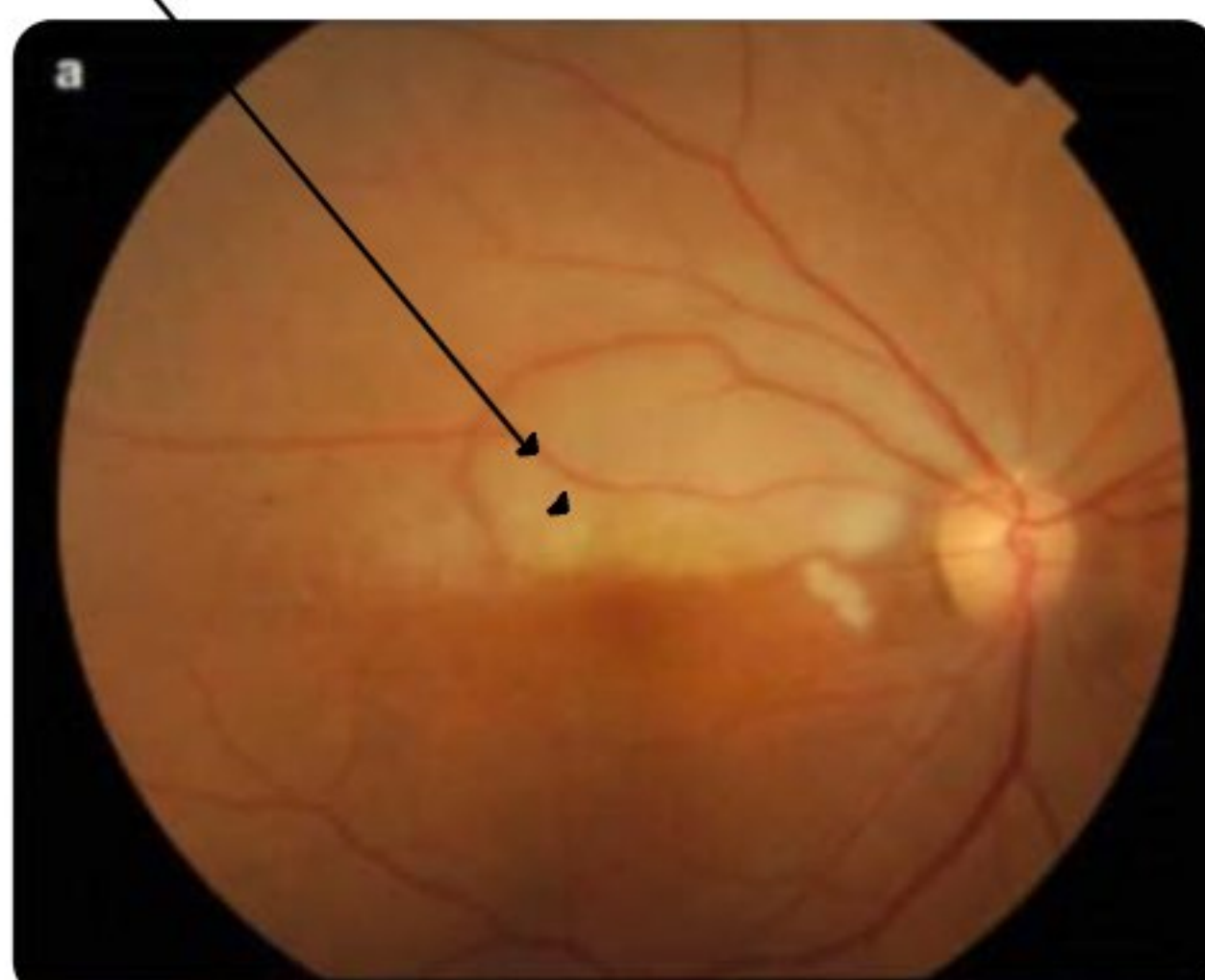
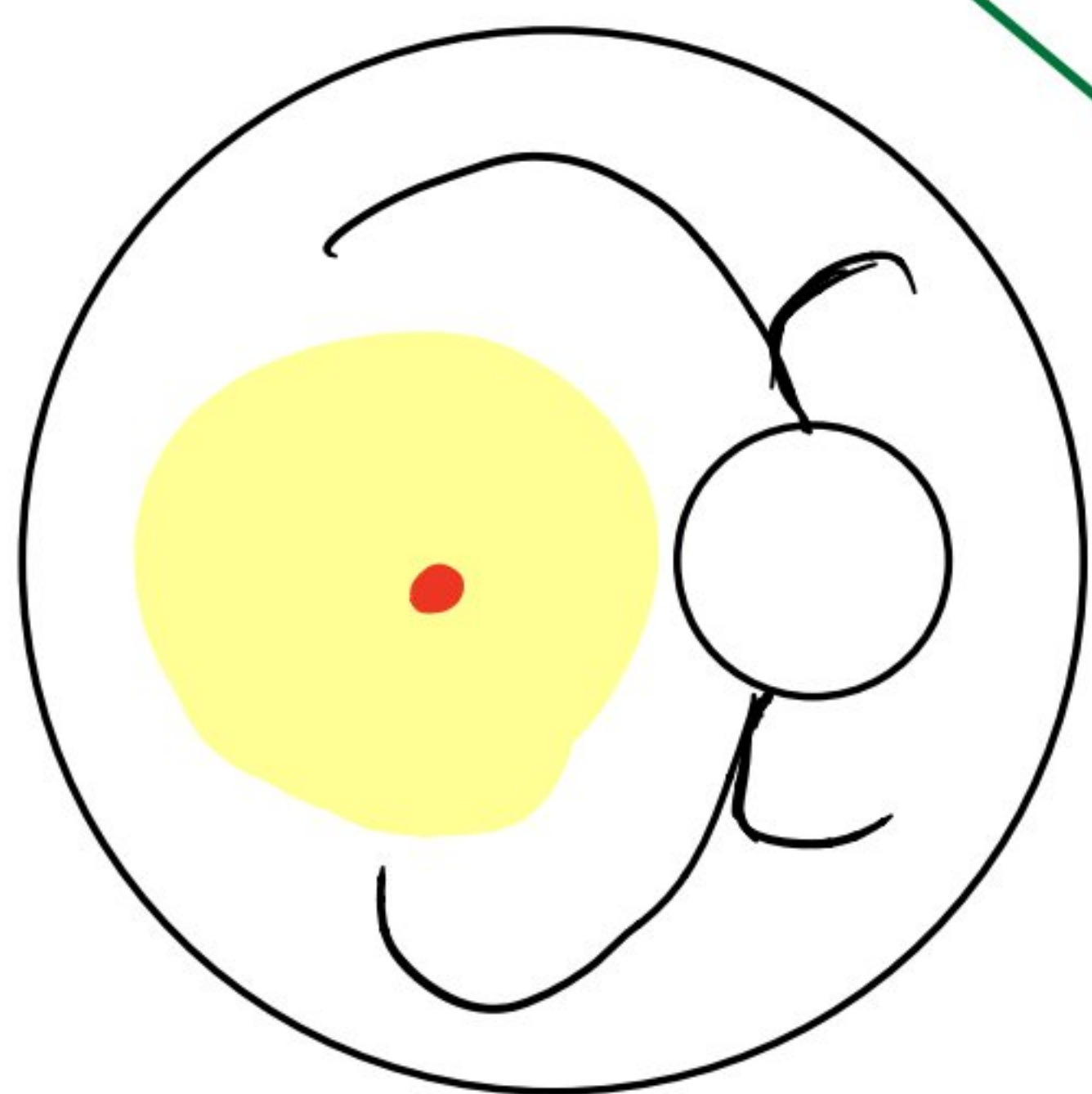
L. Ganglion layer edema

@ Foveola no ganglion layer, so no edema

Cherry red spot

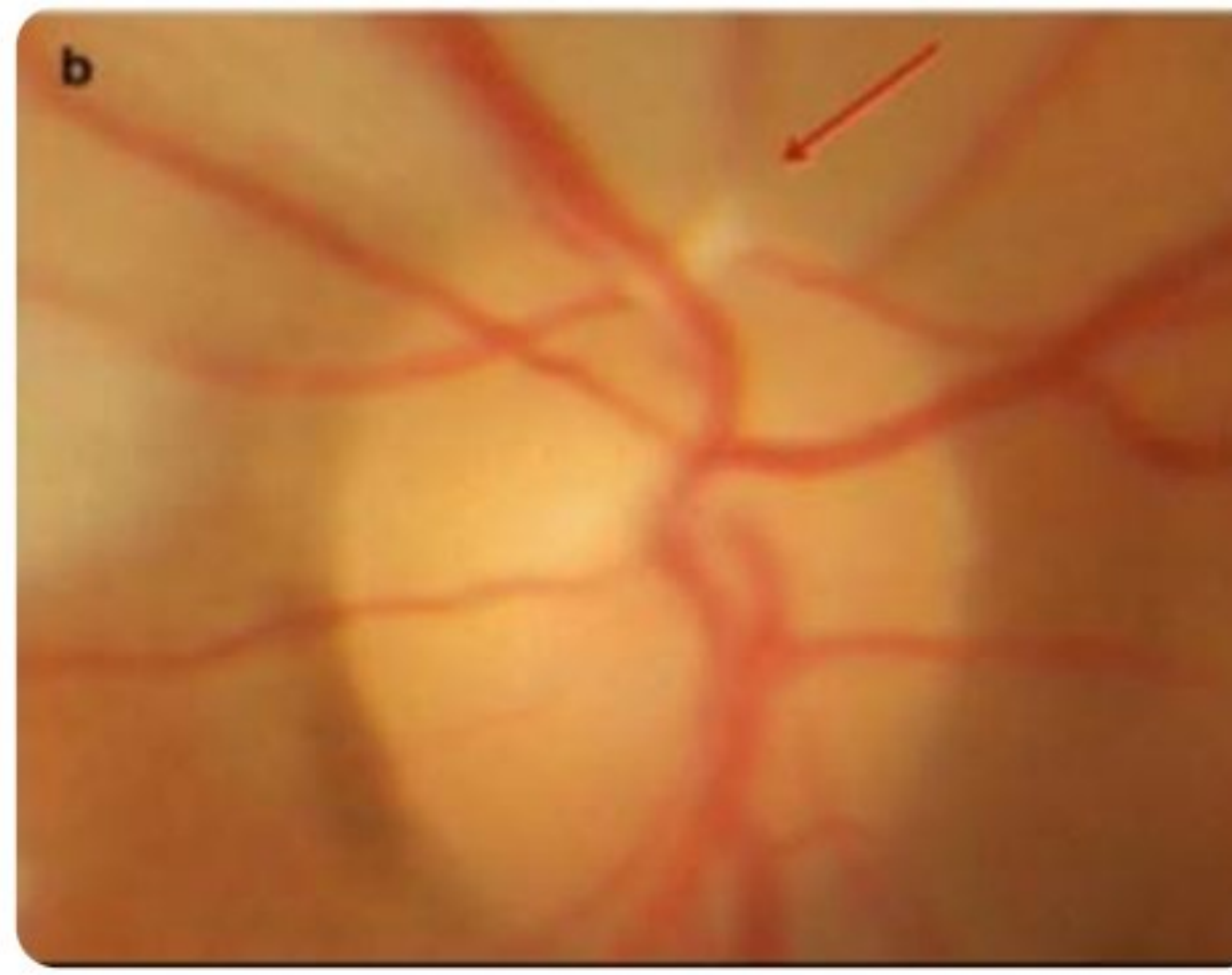
- Arteriolar attenuation
 - Cattle track appearance of Retinal vein
"Box ^{or} carrying"
- NEET PG x2

8. **HOLLNHORST PLAQUE** → cholesterol emboli visualised





CRAO



Investigation: angiography

{ done only
after Mx }

• ↑ Arm - Retina time

• ↑ AV time

• cattle truck appearance & obstruction of artery, dye moves in segment

① ECG

② Carotid doppler } for cause.

Rx: • Remove embolus within **90 minutes**

① Ocular digital massage → may help in dislodge of emboli

② Inhalation of carbogen or amyl nitrate

CO₂ + O₂
5% 95%
it is a vasodilator

③ ↓ IOP ⇒ Acetazolamide / IV Mannitol

⇓

Paracentesis Cinciston in outer chambers
aqueous comes out = ↓ IOP

CAUSES OF CHERRY RED SPOT IN FUNDOSCOPY

CRAO ← Cherry red spot

Blunt Trauma / Berlin Edema ← Trees

Neiman Pick d/s ← Never
Gaucher / GM₁ / GM₂ ← Grow
gangliosidosis

Tay Sachs ← Tall in

Sandhobb's d/s ← Sand &

Metachromatic
Leukodystrophy ← Mud

} Ganglion layer edema

} Storage product deposition in ganglion layer.

Common to Retina : 1, Cherry Red spot
+
2, Intracellular Retinal Edema → Berlin's Edema } d/t Blunt trauma

INNER RETINAL BARRIER BREAK

Eg: Diabetic Retinopathy → Problem of central retinal artery

Risk factors

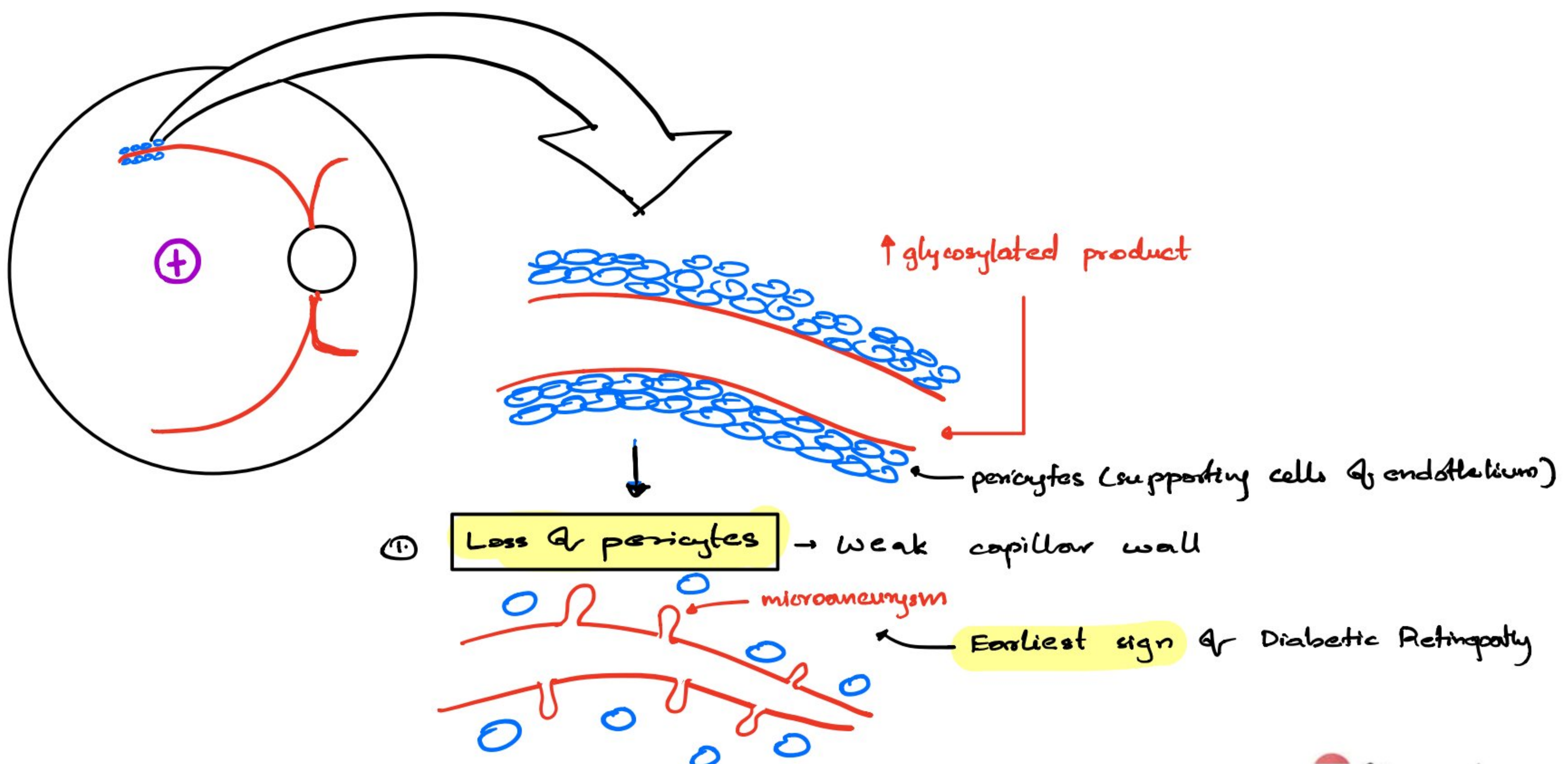
- i) Diabetic duration (Most important) → Duration: long term
→ Glycemic control: Fluctuating
 - ii) Genetics
 - iii) Poor Blood Sugar control
 - iv) High blood pressure
 - v) High cholesterol
 - vi) Smoking
 - vii) Nephropathy
 - viii) Pregnancy in diabetic ♀
- } • BP
• Lipid levels
• Urea / creatinine



Screening for Diabetic Retinopathy : DR are symptom. less in early stage

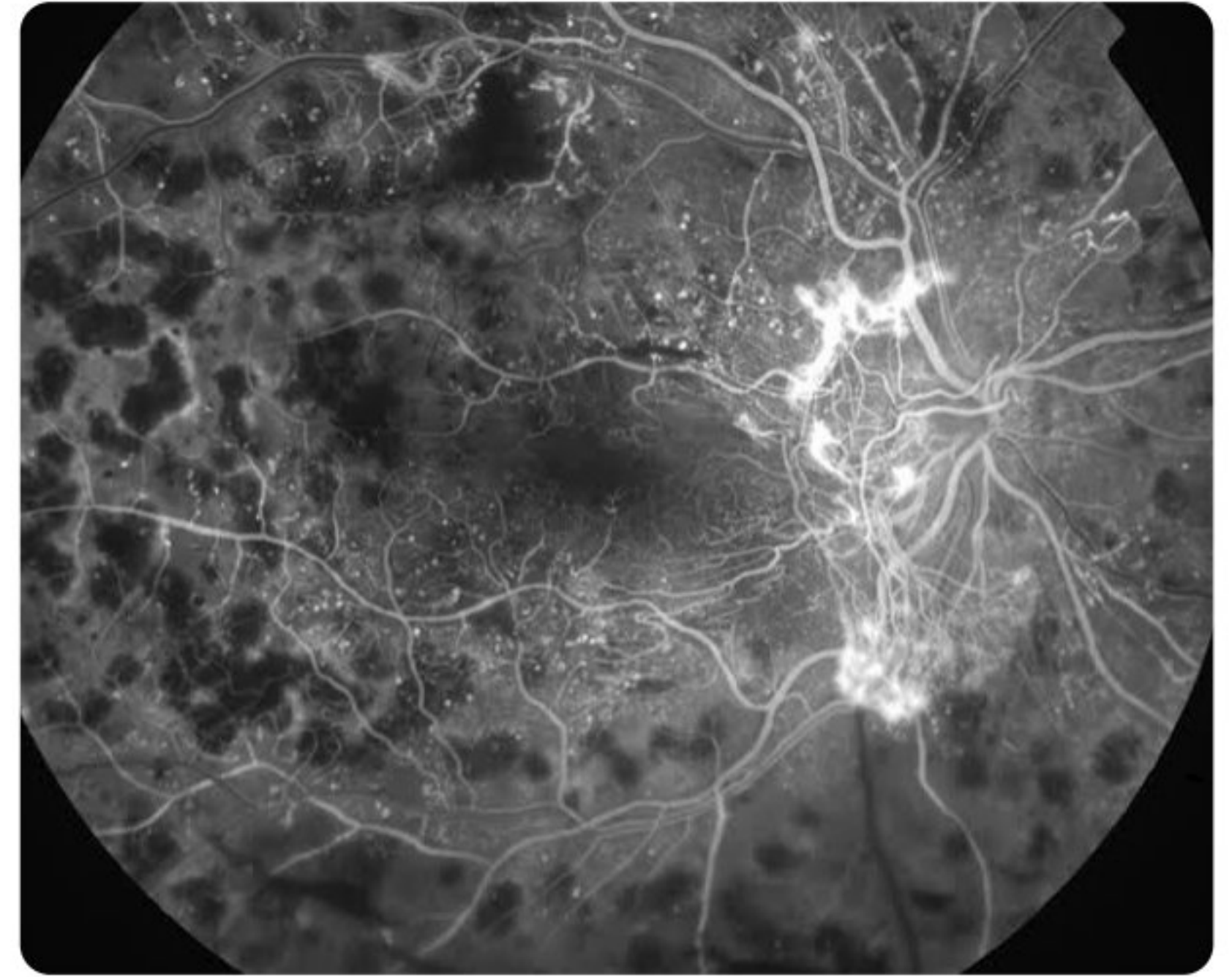
Type 1 DM → 5yr after Δsis of DM $\frac{+}{-}$ Annually } Dilated fundus examination.
Type 2 DM → Immediately upon Δsis of DM → Annually
Diabetes in pregnancy → Every 3 months

Pathogenesis : Microangiopathy.



② Thickening of basement membrane

1st sign $\Rightarrow$ microaneurysms
 $\downarrow$ found
inner nuclear layer
Angiography = Hypofluorescents



Diabetic Retinopathy

SIGNS OF NPDR : 1st sign = microaneurysm

2nd sign = Intraretinal H'mage d/t inner retinal barrier broken

causes : VEGF

- suprt. H'mage $\rightarrow$ Flame H'mage (NFL)
- deep H'mage $\rightarrow$ Dot & Blot H'mage (outer plexiform layer)

• Lipoproteins $\rightarrow$ **Hard exudates**
leak out from inner barrier and reach OPL

• Venous changes :

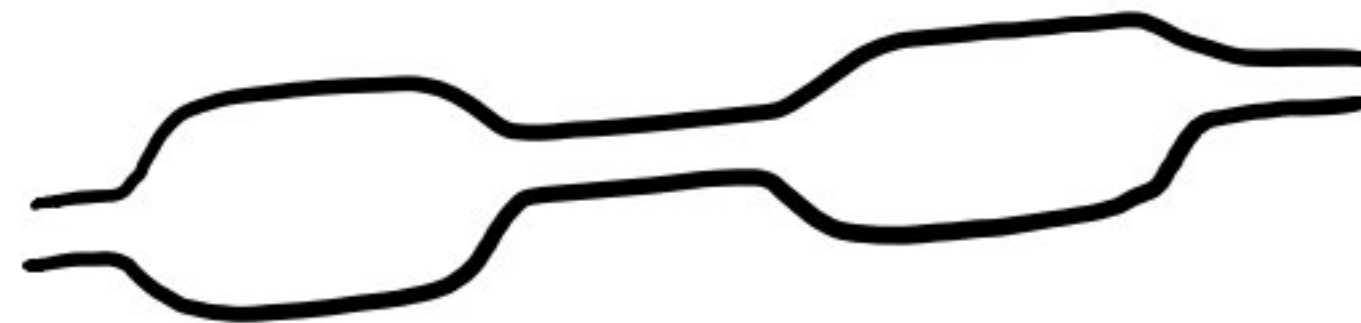
i) Venous bleeding



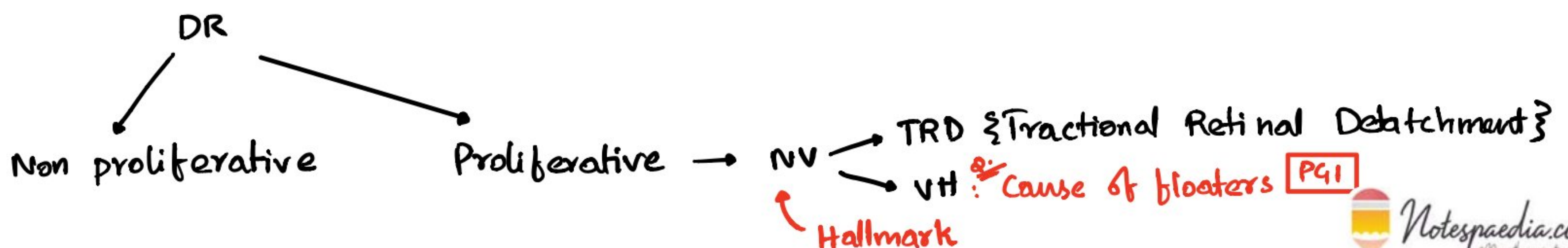
ii) Venous looping



iii) Venous sausageing



Soft exudates $\rightarrow$ **Cotton wool exudate** $\leftarrow$ Sign of significant retinal ischemia
infarction of NFL.

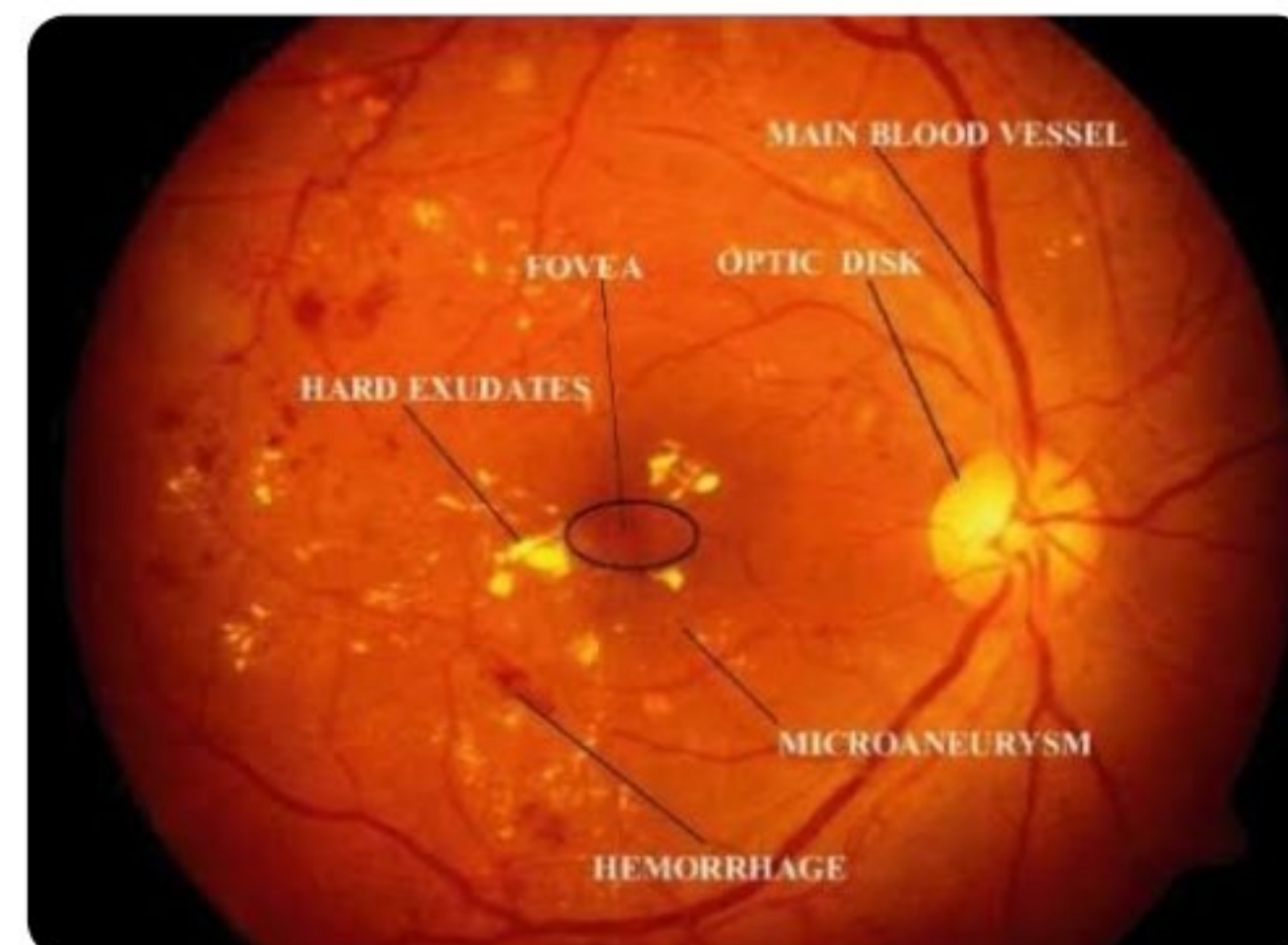
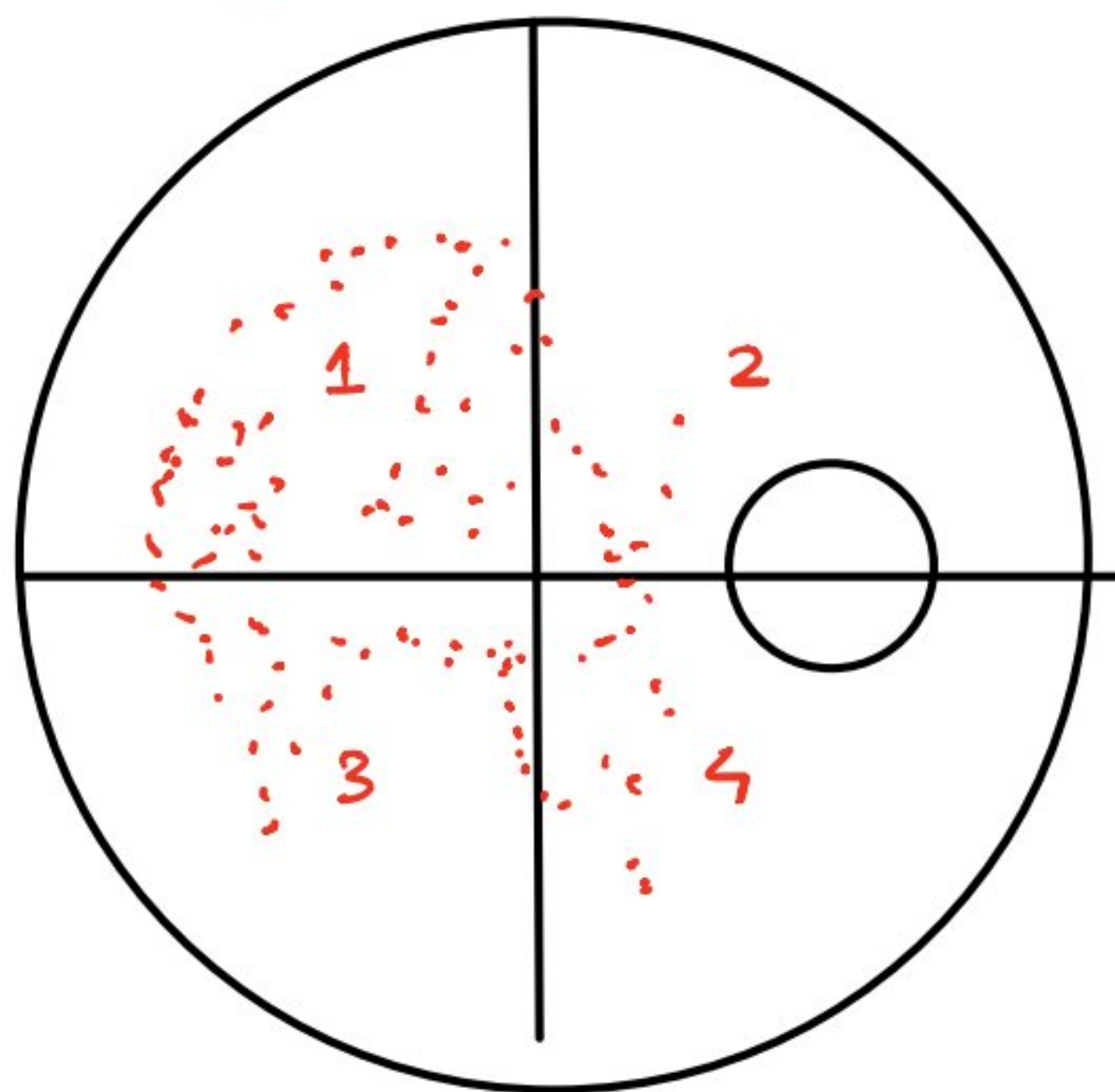


mild → microaneurysm @ venular end
 moderate → H'mage / exudate ≤ 4 quadrants

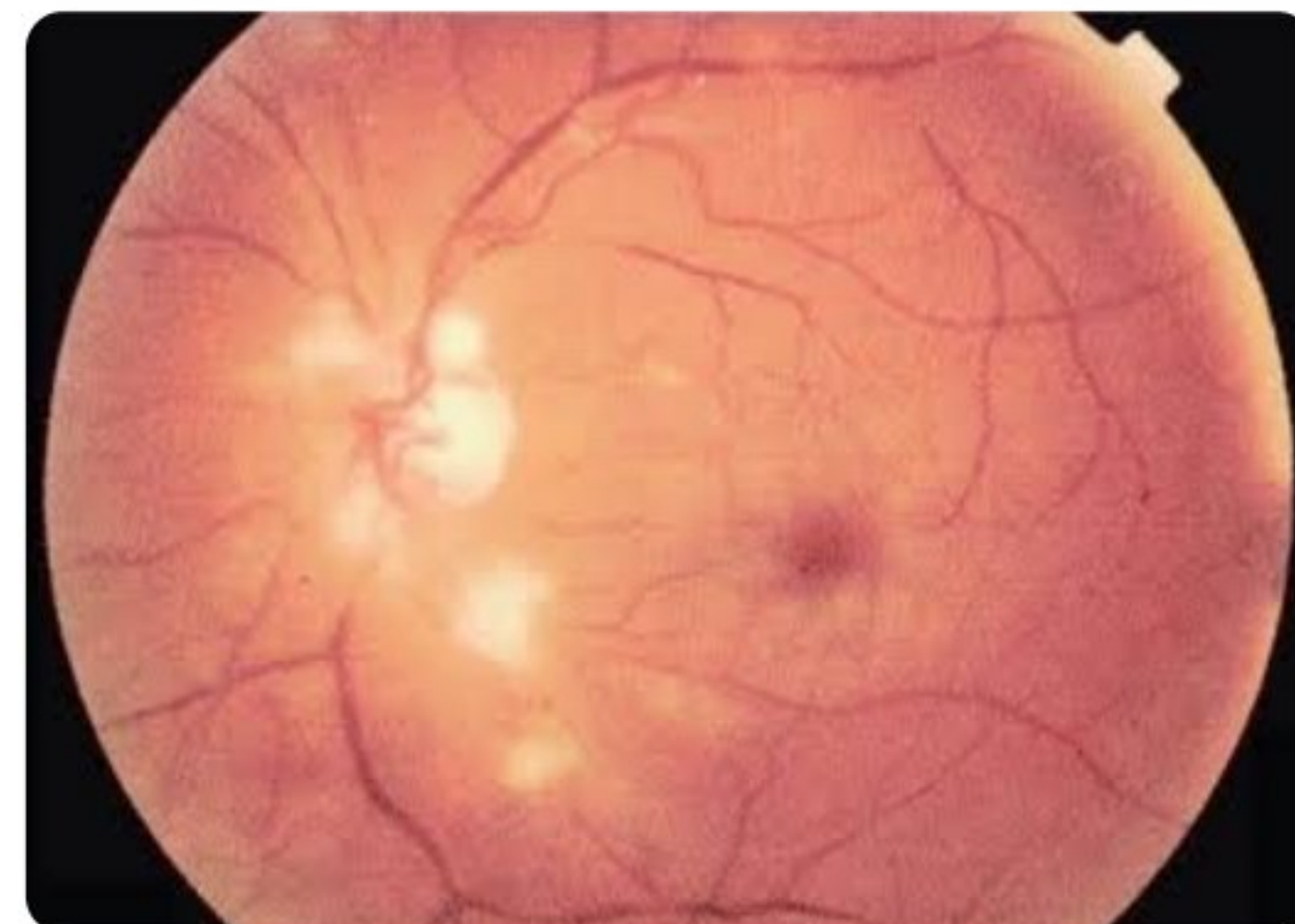
severe → 4:2:1 rule

any one of the rule → Severe NPDA

4 quadrants
 Haemorrhage / exudate

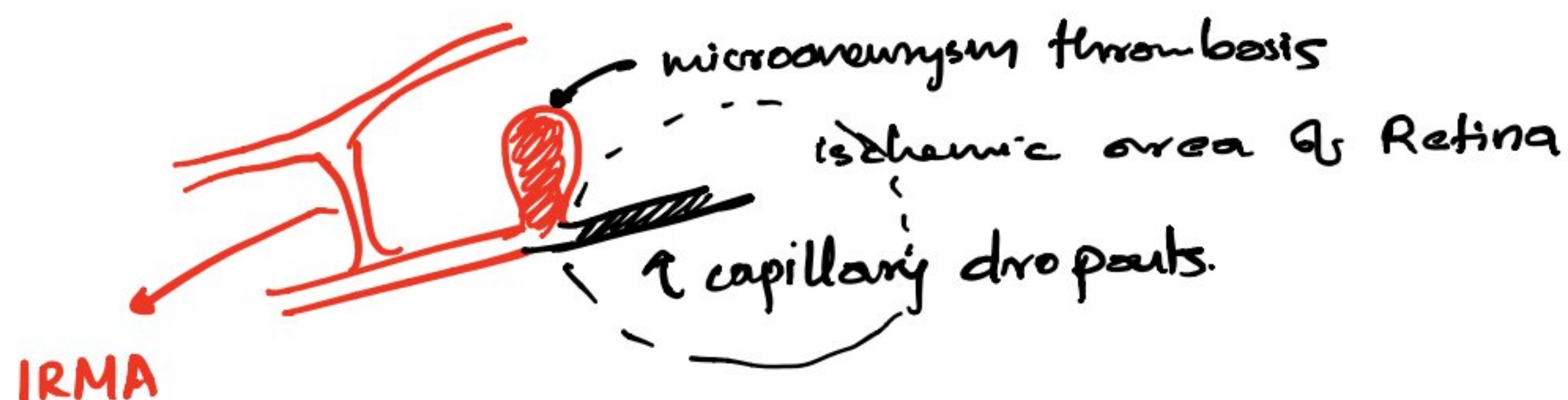


Hard Exudates



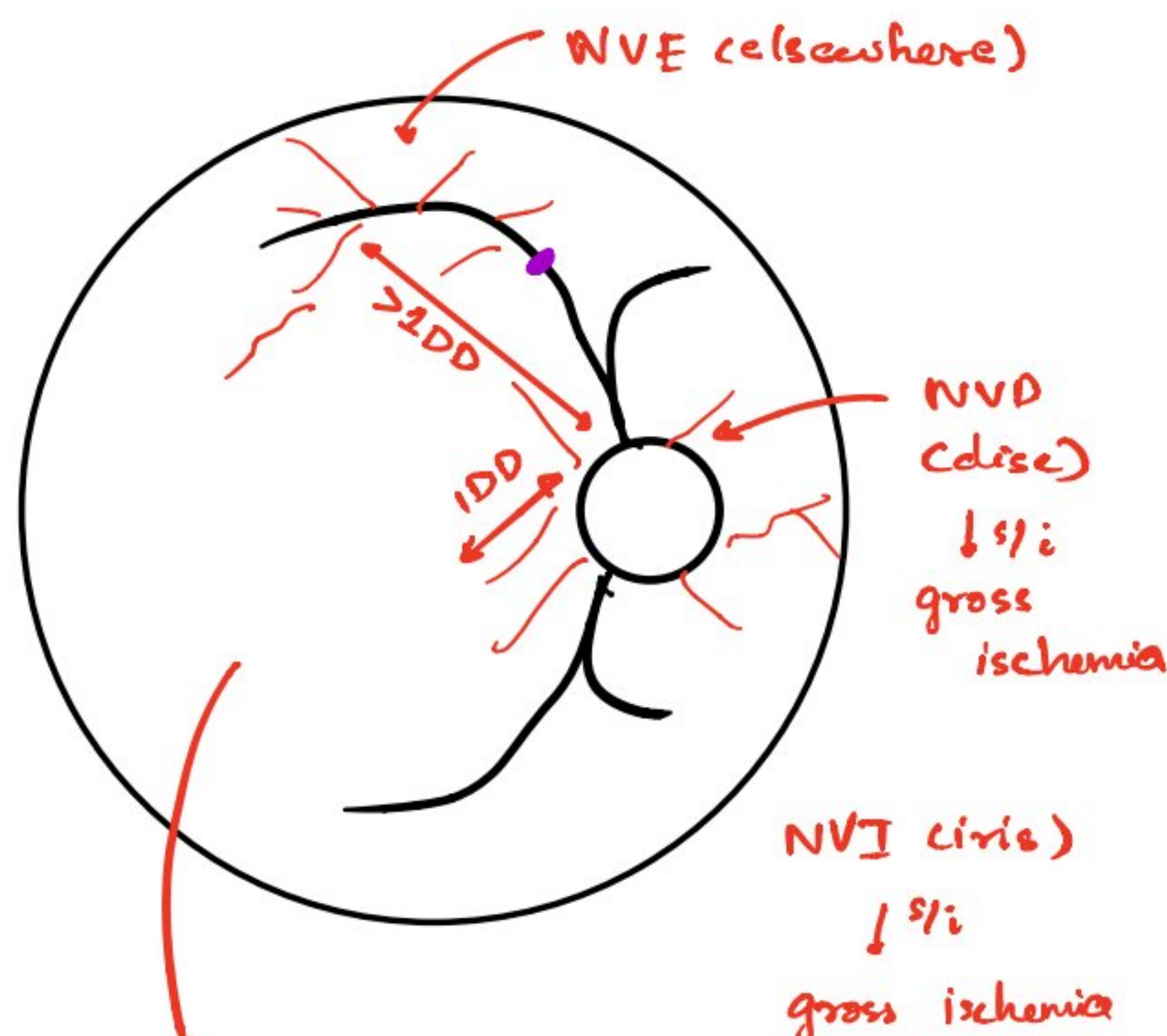
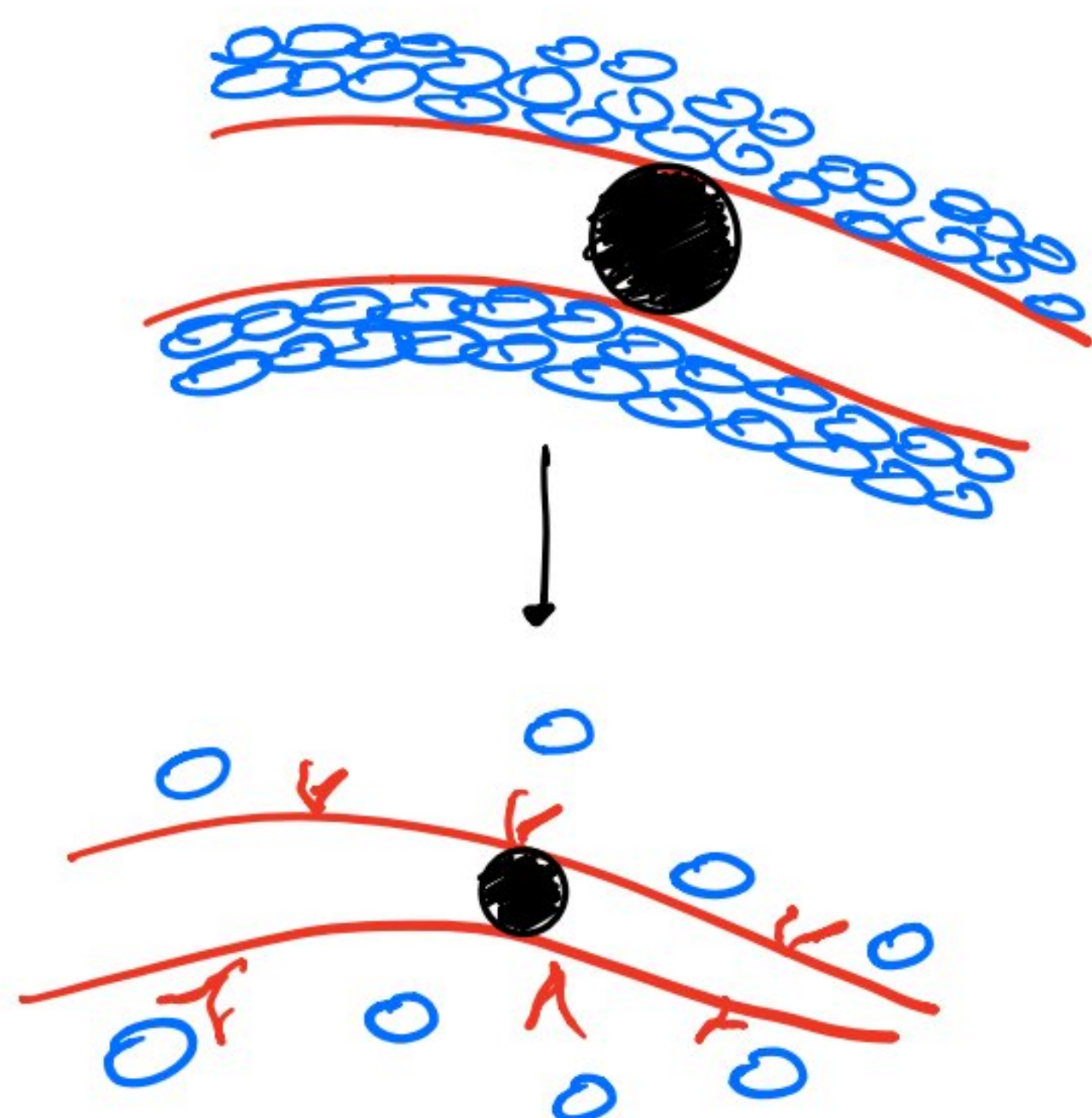
Cotton Wool Spot

• IRMA → Intra retinal microvascular Anomaly:
 Opening up of pre-existing shunts



IRMA

Proliferative : If capillaries get blocked



Hypoxic / ischemic

↑↑ VEGF → new vessel on retina

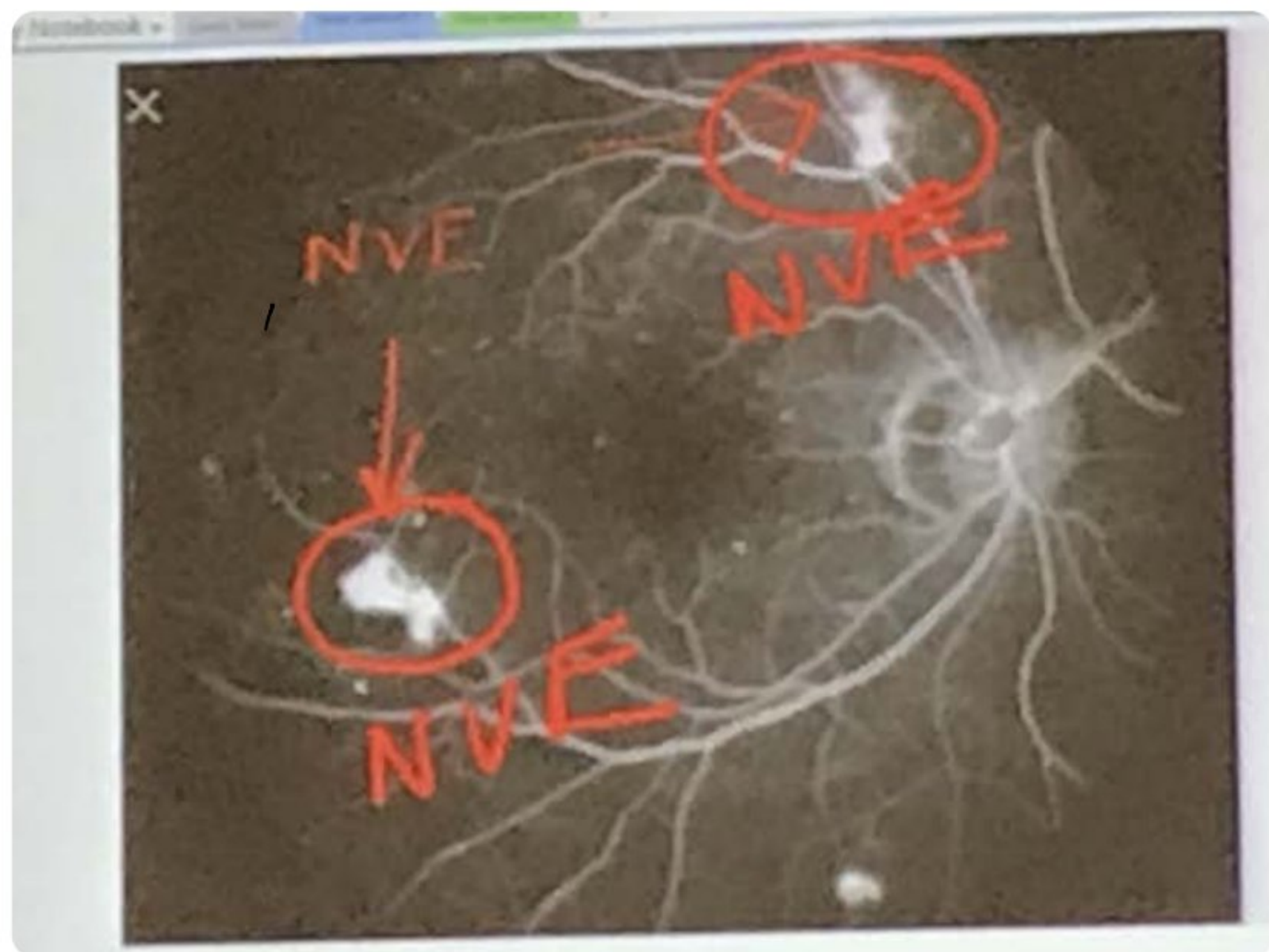
leak blood
 Tractional RD Vascular H'mage

NV - Ab(N)

- 1) Fragile
- 2) Uncontrolled growth
- 3) Fibrous component (pull nearby structures)

Angiography Investigation

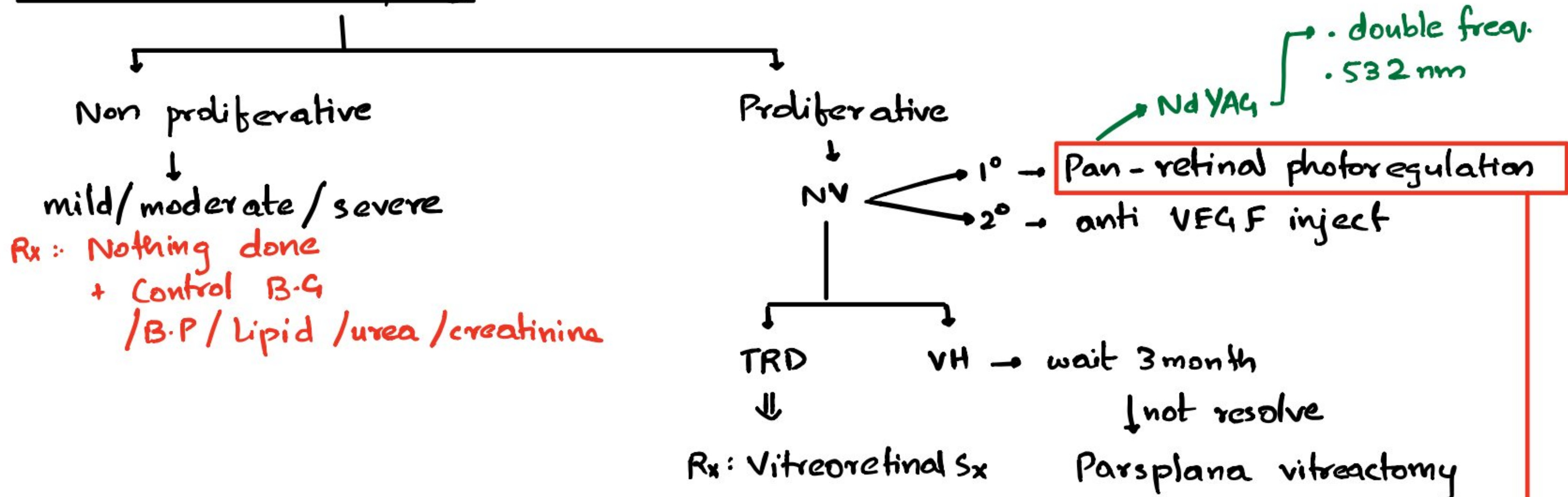
- Micro aneurysm $\Rightarrow$ Hyperfluorescence
- H'mage & exudate causes blocked Hypofluorescence



AIIMS 19

ETDRS $\rightarrow$ Early treatment for diabetic retinopathy study.

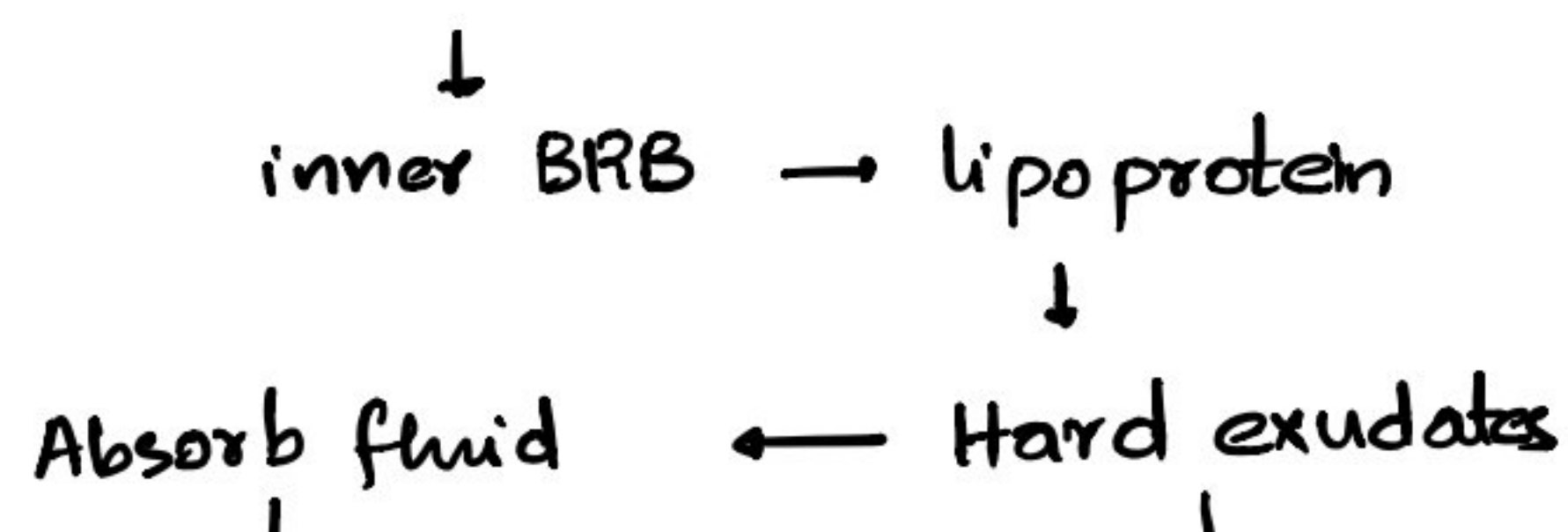
Rx of Diabetic Retinopathy



Follow Up:

mild $\rightarrow$ 1yr
moderate $\rightarrow$ 6month
severe $\rightarrow$ 2-3 month

NPDR + macular edema



? destruction of entire retina except macula 3

$\downarrow$ Hypoxic Retina

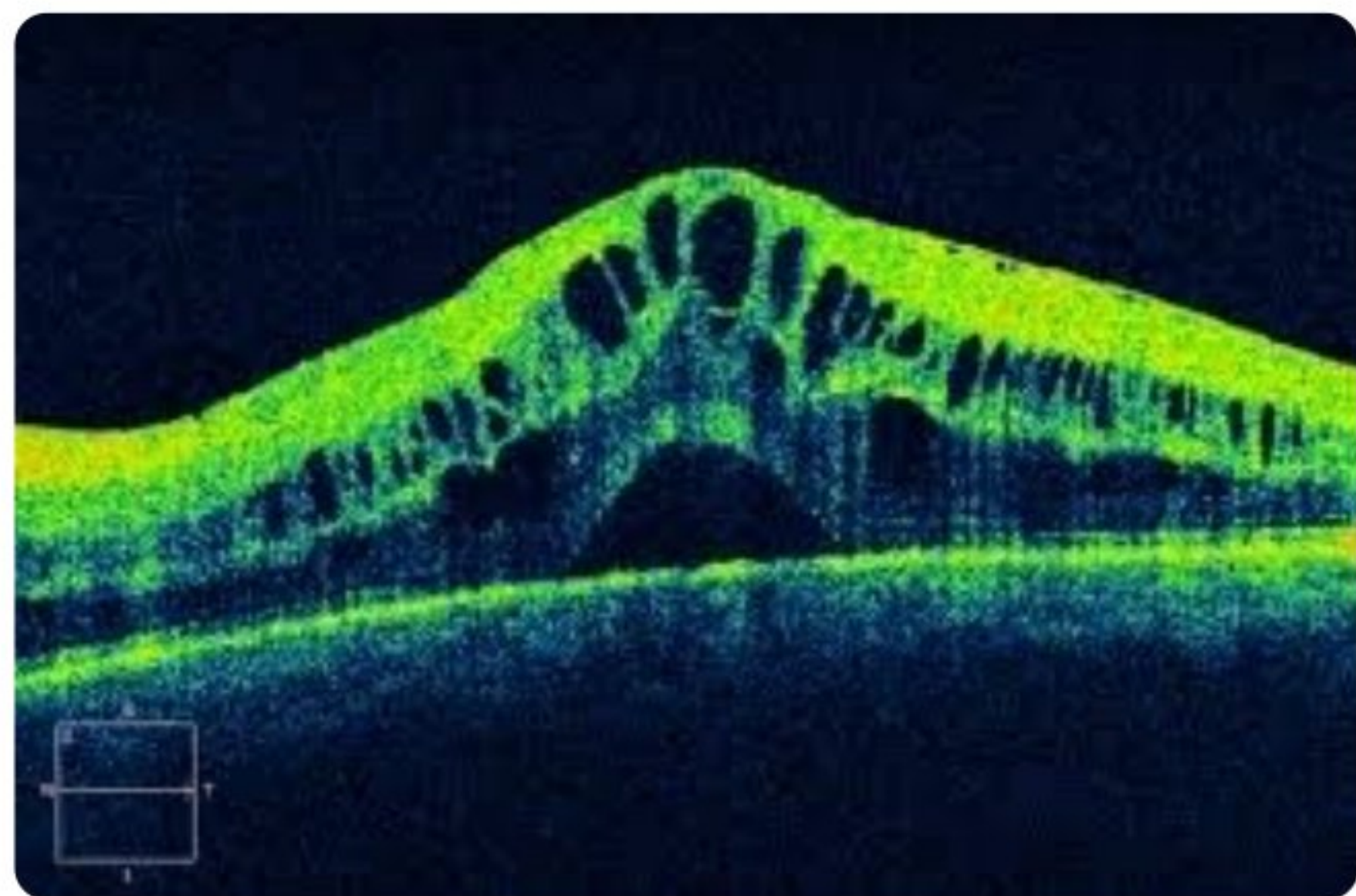
$\downarrow$ VEGF

Anti VEGF : used intravitreally
① BEVACIZUMAB ② RANIZUMAB
③ PEGAPTANIB

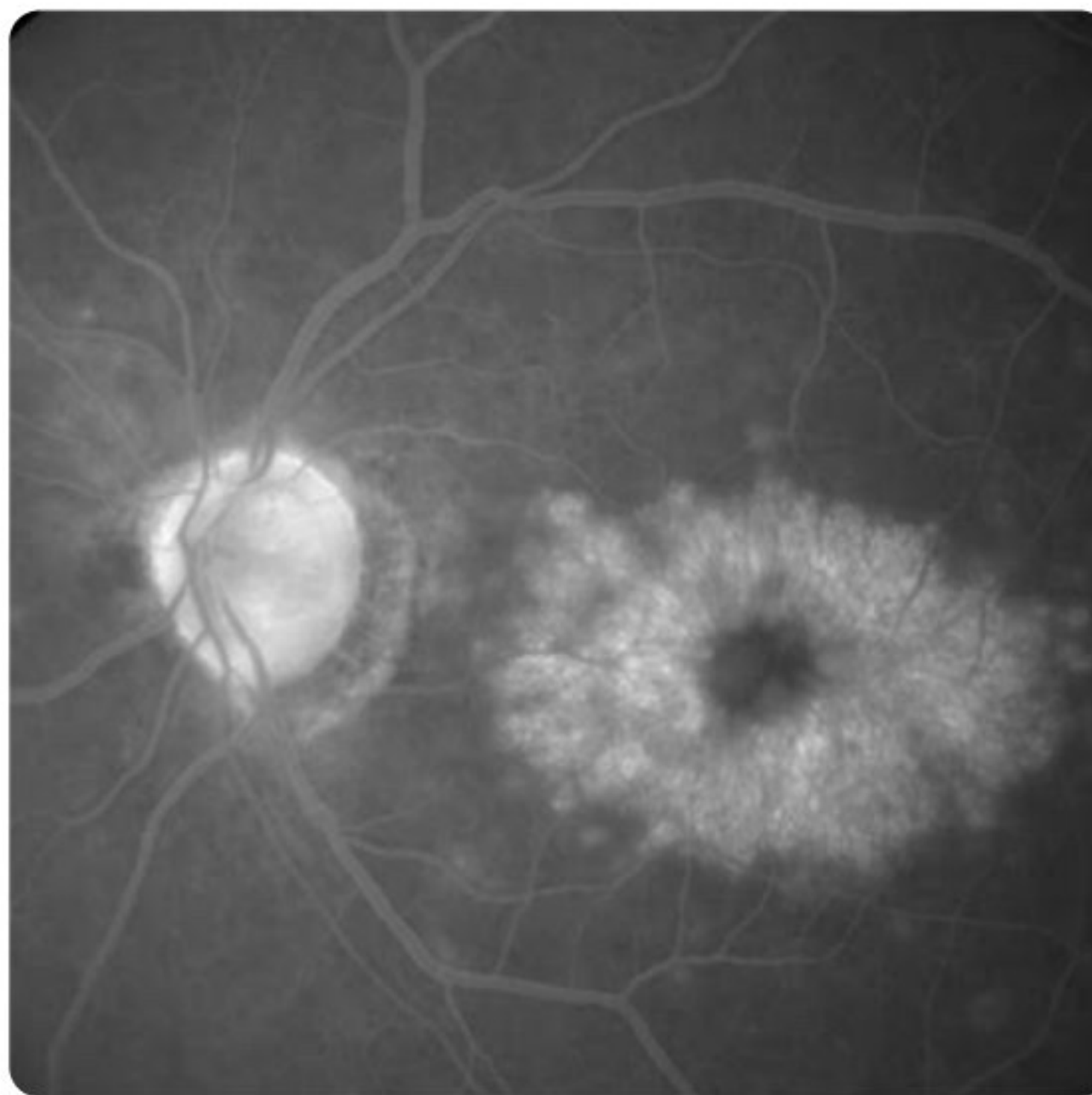
↓
Cystic spaces
↓
Cystoid macular edema

↓
OPL
↓
↑ Macular Thickness

Inx : OCT
Angiography



Cystoid Macular Edema



• Petalloid/flower petal arrangement of CME
• Dye → OPL → Macula

DRUGS causing cystoid Macular Edema

- 1) PG analogues
- 2) Epinephrine
- 3) Dipivefrine {aphakic eye}
- 4) Tamoxifen
- 5) Docetaxel
- 6) Thiazolidine diones
- 7) Miacin {pseudo CME}

Rx: Intravitreal injection of anti VEGF

- | | |
|-------------------------|----------|
| i) Bevacizumab → 1.25mg | } VEGF ⓐ |
| ii) Ranibizumab → 0.5mg | |
| iii) Aflibercept → 2mg | |
- ↳ VEGF trap

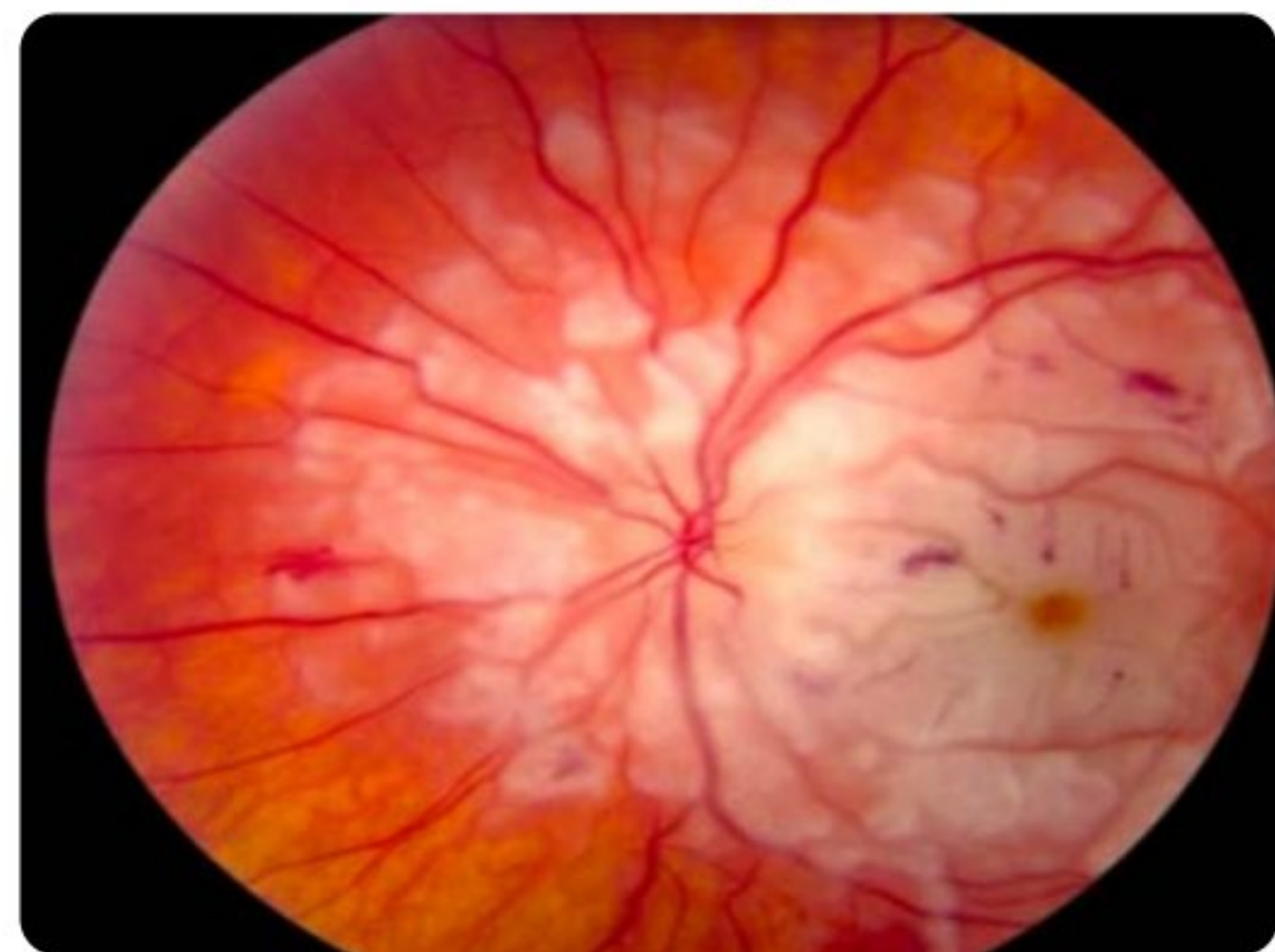
PURTSCHER'S RETINOPATHY

- Multiple cotton wool spots around the disc
- associated with severe head injury
- also s/i pancreatitis and chest trauma d/t fat/air embolism.

No effective treatment

PURTSCHER'S RETINOPATHY

⚡ The following image shows the picture of the fundus in a patient taken 48 hours after sustaining severe head trauma. What is the diagnosis?



- a) Hypertensive retinopathy
 b) Diabetic retinopathy
 ✓ c) Purtscher's retinopathy
 d) Central retinal vein